Cardiac Dynamics: Mathematical Study of Cardiomyocyte Models





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Resumen

El análisis de sistemas dinámicos se ha convertido en una disciplina fundamental para el estudio de múltiples problemas de la vida real. El objetivo es modelizar fenómenos para comprender su progreso a lo largo del tiempo. Una adecuada modelización matemática permite extraer conclusiones sobre el comportamiento de un sistema con gran utilidad en áreas que requieren un análisis cualitativo y cuantitativo de la evolución de las variables dinámicas. Esto es de gran importancia en campos muy variados como la física, la economía o la biología, donde los modelos se utilizan para explicar los cambios que experimentan diversas magnitudes dinámicas y para realizar predicciones.

Las biomatemáticas son un área de las matemáticas que ha experimentado un notable crecimiento en los últimos años. Se encarga del estudio de métodos para entender los procesos dinámicos de la biología a través de modelos matemáticos. Se trata de una rama multidisciplinaria donde convergen médicos, físicos, biólogos, estadísticos y otros perfiles que contribuyen al entendimiento de los sistemas biológicos y posibles aplicaciones del conocimiento adquirido. Esto incluye, por ejemplo, el desarrollo de fármacos específicos mediante modelos muy precisos, la secuenciación del ADN o cualquier simulación de un sistema biológico mediante técnicas numéricas. El desarrollo en esta área se ha visto favorecido por los recientes avances tecnológicos y el incremento en la potencia computacional para llevar a cabo análisis de sistemas complejos.

En este trabajo, se presenta una serie de fundamentos matemáticos y técnicas con aplicación en sistemas biológicos, así como una descripción y análisis de algunos modelos concretos de cardiomiocitos: las células del músculo cardiaco.

El **capítulo 1** (*Introduction and Motivation*) es una breve introducción al funcionamiento de los sistemas de propagación eléctrica del corazón. Se describen brevemente las partes del corazón involucradas en el sistema de conducción de la electricidad. También se mencionan las fases que experimenta el potencial a través de las membranas de los cardiomiocitos entre la despolarización inicial cuando se produce un rápido aumento del voltaje y la repolarización final cuando se produce un descenso hasta el potencial de base. Este proceso de rápidos cambios en el potencial de la célula es lo que se conoce como potencial de acción. Además, se expone la motivación que existe detrás del estudio de modelos del corazón como la explicación de anomalías y el desarrollo de posibles tratamientos.

El **capítulo 2** (*Dynamical Systems Theory*) está dedicado a proporcionar una explicación autocontenida de la teoría de sistemas dinámicos que será empleada más adelante. Se pondrá un especial énfasis en aquellas técnicas y elementos que tienen una relación directa con la descripción de ciertos sistemas biológicos como cardiomiocitos o neuronas con comportamiento excitable. Se presentarán los llamados sistemas *fast-slow* que modelan simultáneamente las diferentes escalas de tiempo en las que a menudo evolucionan las variables biológicas de un sistema. Dependiendo de la escala de tiempo (escala rápida con tiempo t o escala lenta con tiempo $\tau = \varepsilon t$ y $0 < \varepsilon \ll 1$), se puede escribir un sistema *fast-slow* de dos formas:

$$\begin{cases} x' = f(x,y,\varepsilon) \\ y' = \varepsilon g(x,y,\varepsilon) \end{cases} \iff \begin{cases} \varepsilon \dot{x} = f(x,y,\varepsilon) \\ \dot{y} = g(x,y,\varepsilon) \end{cases}$$

Por ejemplo, esto es útil para modelizar paralelamente rápidos procesos de excitación y propagación de electricidad y lentos procesos de relajación o apertura de canales de iones que portan carga eléctrica.
Posteriormente, se define la variedad crítica de un sistema *fast-slow* como el conjunto de puntos donde $f(x,y,0) = 0$ y proporciona información sobre el sistema en el límite $\varepsilon \to 0$. Tras ello, se introducen algunos conceptos fundamentales sobre el análisis de un sistema a través de sus puntos de equilibrio y, en particular, a través de los equilibrios hiperbólicos donde el jacobiano del sistema no tiene valores propios con parte real nula y se pueden clasificar como atractores, repulsores o puntos de silla. Estos conceptos se extienden a subconjuntos de la variedad crítica y se introduce el concepto de subconjunto normalmente hiperbólico donde el jacobiano con respecto a las variables rápidas del sistema no tiene valores propios con parte real nula. Los subespacios estables e inestables se construyen con los vectores propios asociados a valores propios del jacobiano del sistema linealizado. También se definen variedades estables e inestables de puntos de equilibrio como los conjuntos de puntos tal que las soluciones que pasan por ellos tienden asintóticamente al punto de equilibrio con el tiempo fluyendo hacia adelante o hacia atrás, respectivamente. El teorema de existencia de variedades locales estables e inestables garantiza la existencia de aproximaciones locales tangentes a los subespacios estables e inestables y compartiendo algunas propiedades. Asimismo, se definen las variedades locales asociadas a conjuntos normalmente hiperbólicos como la unión de las variedades locales asociadas a cada uno de los puntos del conjunto.
Con estas bases, se puede introducir uno de los teoremas más importantes de este trabajo y de la teoría geométrica de perturbaciones singulares: el Teorema (o Teoremas) de Fenichel. Bajo ciertas hipótesis, garantiza la persistencia de la estructura topológica y el comportamiento de la variedad crítica calculada en el límite $\varepsilon \to 0$ cuando pasamos a estudiar el sistema real fuera del límite, así como la persistencia de las correspondientes variedades locales construidas a partir de dicha variedad crítica. Además, proporciona resultados acerca de la existencia de las nuevas variedades reales a las que la variedad crítica se perturba y acerca de la convergencia y la diferenciabilidad. Por último, se presenta también un fenómeno especial de los sistemas *fast-slow* relacionado con conceptos anteriores: los *canards*. Se trata de soluciones del sistema en ciertos intervalos donde la trayectoria se mantiene muy cercana a una rama repulsora de la variedad lenta asociada a una subvariedad normalmente hiperbólica de la variedad crítica. Habitualmente, se habla también de *canards* maximales consistentes en soluciones que se encuentran en la intersección de una rama repulsora y una rama atractora de la variedad lenta.

En el **capítulo 3** (*Application to Cardiomyocyte Models*), se procede al estudio de modelos matemáticos concretos de cardiomiocitos. Al final del capítulo anterior se menciona brevemente uno de los primeros modelos de cardiomiocitos diseñados con el objetivo de capturar las propiedades de excitación: el Modelo de FitzHugh-Nagumo. Este modelo se contruye sobre la base del oscilador de Van der Pol que es el ejemplo prototípico de presencia de *canards*. No obstante, este capítulo se centra en el estudio de otros dos modelos ligeramente más avanzados: el Modelo de Karma en 2 dimensiones y el Modelo Reducido de Luo-Rudy en 3 dimensiones. Ambos modelos presentan características y conclusiones únicas y dan una idea clara del modo en que los modelos se contruyen para intentar modelizar los procesos físicos de la realidad. Para ambos modelos, se aplican los conceptos estudiados en el capítulo anterior.
El Modelo de Karma viene representado por un sistema de dos ecuaciones diferenciales en derivadas parciales. No obstante, en este trabajo, no se analizará la contribución de los fenómenos de difusión a lo largo de las dimensiones espaciales y se procederá al análisis de la versión del modelo como sistema de ecuaciones diferenciales ordinarias. El modelo presenta una ecuación rápida para la evolución del voltaje o potencial eléctrico a través de la membrana de un cardiomiocito (variable E) y una ecuación lenta para cuantificar el grado de apertura de los canales de iones que fluyen y generan cambios en dicho voltaje (variable n). Se explica el significado físico detrás de los parámetros empleados y diversos detalles de interés sobre la construcción del modelo final. Después, se demuestra la naturaleza *fast-slow* de este

modelo y se procede a la obtención de magnitudes estudiadas en el capítulo anterior. Se realizan cálculos sobre la variedad crítica y los puntos de equilibrio. Se realiza también una simulación numérica para observar el diagrama de fases del sistema y la evolución temporal de cada variable en una órbita típica. Una vez que se tiene una buena idea de la naturaleza del sistema, se aplica la teoría de Fenichel para confirmar la perturbación de las órbitas a partir de las órbitas candidatas en el límite $\varepsilon \to 0$. Además, se aplica el teorema de Poincaré-Bendixson en una región invariante para demostrar la existencia de una conexión heteroclínica en el sistema completo.
El Modelo Reducido de Luo-Rudy viene representado por un sistema de tres ecuaciones diferenciales ordinarias. De forma similar al Modelo de Karma, se tiene una ecuación que representa la dinámica del voltaje o potencial transmembrana (variable V) que se modelizará como una ecuación rápida. Sin embargo, ahora se tienen hasta dos ecuaciones más para representar a los canales de iones de calcio (variable f) y potasio (variable x) que se modelizarán como ecuaciones lentas. Se analizan las ecuaciones del modelo con la correspondiente explicación de los parámetros y funciones empleadas. Después, se plantea la formulación del modelo como sistema *fast-slow*. Se calcula la variedad crítica, las líneas que separan sus diferentes comportamientos y se aplica la teoría de Fenichel para demostrar la persistencia de las variedades en el sistema completo con las propiedades vistas anteriormente. Luego, se realizan simulaciones numéricas para varias combinaciones de parámetros variando la conductividad asociada al canal del potasio. En todos los casos se observa la presencia de una órbita periódica a la que acaban cayendo las trayectorias y se dan pulsos periódicos que ilustran las diferentes fases por las que pasa el potencial de un cardiomiocito con comportamiento excitable. No obstante, se observan diferencias sutiles en la forma de las órbitas que están relacionadas con procesos que podrían generar arritmias. Esto se detalla posteriomente con ilustraciones completas de los elementos calculados y la presencia de *canards*, así como sus implicaciones biofísicas.

El **capítulo 4** (*Conclusion and Further Work*) es una conclusión del material expuesto en el trabajo y los resultados fundamentales que se han obtenido. Se revisan las observaciones de los Modelos de Karma y Luo-Rudy y se exponen posibles líneas de trabajo futuro. Se hace hincapié en la importancia del estudio de sistemas dinámicos de baja dimensión para llegar a comprender las propiedades de sistemas más complejos y saber cómo construirlos. En la actualidad, se siguen publicando resultados sobre modelos de dimensión baja por la importancia de algunos resultados cualitativos que se obtienen. No obstante, el desarrollo de herramientas computacionales suficientemente potentes permite la obtención de resultados cuantitativos más precisos a través de modelos más complejos. Se expone brevemente el ejemplo del llamado Modelo de Sato con 27 variables dinámicas y 177 parámetros para modelar los complejos procesos biofísicos del corazón. Se han propuesto numerosos modelos con muy diversos grados de complejidad y su estudio es fundamental para comprender el origen de enfermedades, desarrollar posibles tratamientos químicos y dar explicación a anomalías según las variables y parámetros que el modelo contemple. Esto crea un vínculo entre las matemáticas y otras ramas de la ciencia, demostrando el potencial de las matemáticas como disciplina aplicada.

Además, se proporciona un **anexo** (*APPENDIX*) con material complementario y algunos comentarios y análisis adicionales. El primer punto es un sumario de los principales enunciados de los teoremas de Fenichel. Después, se presenta un breve inciso sobre el efecto de añadir una corriente adicional en la ecuación del potencial eléctrico en el Modelo de Karma viendo los cambios que produce en algunos de los objetos calculados para el caso de nula corriente externa en el capítulo 3. Después, se plantea esto mismo para el Modelo de Luo-Rudy, así como un análisis de conductividades del canal de potasio aún menores que las analizadas en el trabajo y un breve resumen de algunas bifurcaciones que podemos encontrar en el sistema. Posteriormente, se dedica una breve sección a la explicación del método de resolución numérica de ecuaciones diferenciales ordinarias utilizado (Runge-Kutta de orden 4). Finalmente, se adjunta el código de Python para simulaciones y gráficas que han aparecido en el trabajo.

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Chapter 1

Introduction and Motivation

Essentially, a mammalian heart is divided into four main chambers. On the one hand, the two upper chambers are called the *left atrium* and the *right atrium*. The atria receive the blood that enters the heart. Also, a wall called the *interatrial septum* separates the two atria. On the other hand, the two lower chambers are called the *left ventricle* and the *right ventricle*. These chambers pump blood out of the heart. The *interventricular septum* is a wall that separates the two ventricles.
Nevertheless, in this thesis, we will focus on the electrical activity occurring in the heart and, in particular, across **cardiomyocytes**, the cardiac muscle cells. These cells contract, get excited and conduct **action potentials**. The action potential is an electrical response that represents a short change in the membrane potential across the cell membrane of heart cells. The following figure depicts some of the most relevant parts of the electrical activity flow process.

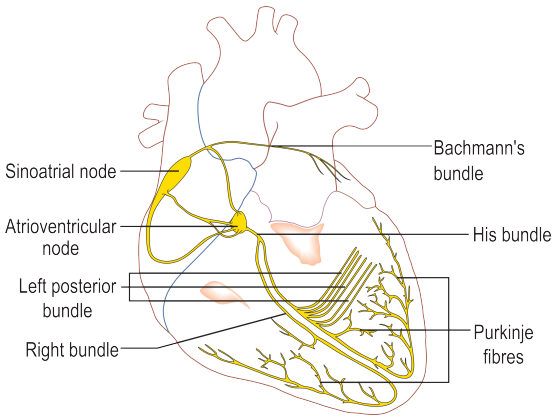


Figure 1.1: Electrical conduction system of the heart (extracted from [1]).

The electrical activity conduction system of the heart starts in the *sinoatrial node*, i.e., a group of cells in the wall of the right atrium. The electrical activity causes atrial contraction. Then, the electrical signal is conducted towards the *atrioventricular node*, i.e., a group of cells in the interatrial septum slowing down conduction of the action potential. This gives time to the ventricles to fill with blood. After that, the electrical signal goes through a bundle of fibers called the *His bundle* between the ventricles. Finally, it reaches the *Purkinje fibers*, causing ventricular contraction. When stimulated, cardiomyocytes in the ventricles suddenly increase their transmembrane voltage (*depolarization*), then partially decrease it (*transient repolarization*) and this is followed by a *plateau* phase where the voltage remains approximately constant. Finally, the transmembrane voltage decreases (*repolarization*) and returns to the resting potential level until the next stimulus.
A healthy heart will present a normal conduction of electrical activity and it will coordinate the excitation and contraction of the heart without major problems. On the contrary, anomalies in excitability or contractility can lead to certain pathologies such as **arrhythmias** when the heartbeat is abnormal. This motivates the study of the evolution of some magnitudes because it can help us understand mathematically the behavior of cardiomyocytes and their excitability properties. After a mathematical study, phenomena can be translated into biological language. This allows to direct scientific research towards more specific regions where the mathematical study shows special characteristics. Also, this may help to suggest more precise clinical treatments in terms of chemical proportions, understanding the flow of ionic currents across the cell membranes of cardiomyocytes and how this affects the voltage in the heart cells. According to the physical and medical characteristics of the patients, this can help develop personalized diagnoses and treatments of heart conditions.

Chapter 2

Dynamical Systems Theory

In this chapter, we will describe the most relevant concepts of dynamical systems theory that will be needed in the next chapter when we analyze specific models. We will discuss a special kind of systems called *fast-slow* that appear frequently when modeling biological systems such as cardiomyocytes or neurons with excitable behavior. From the framework of fast-slow systems, we will see the importance of the study of certain equilibrium points and manifolds. We will also study the main ideas of geometric singular perturbation theory, i.e., *Fenichel Theory*. Finally, we will examine *canards*, which is a special phenomenon that occurs in fast-slow systems.

2.1 Fast-Slow Systems

In order to model the two different time scales on which the variables evolve, let us introduce a particular type of systems of differential equations:

Definition 2.1. Let $x \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$ be state space variables evolving over time. Let ε be a small real parameter, i.e., $0 < \varepsilon \ll 1$. Let $f : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^m$ and $g : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ be smooth functions. A system of differential equations of the form

$$\begin{cases} x' = f(x,y,\varepsilon), \\ y' = \varepsilon g(x,y,\varepsilon) \end{cases} \tag{2.1}$$

is called an *(m,n)-fast-slow system*. That is, a system of differential equations with m fast equations and n slow equations. Its associated vector field is $X_\varepsilon(x,y) = (f(x,y,\varepsilon), \varepsilon g(x,y,\varepsilon))$. The variables $x_1, \ldots, x_m$ contained in the vector x will be called *fast* and the variables $y_1, \ldots, y_n$ contained in the vector y will be called *slow*.

As used in Definition 2.1, we will denote the derivative with respect to time t by the *prime* notation, i.e., $x' := \frac{dx}{dt}$ and $y' := \frac{dy}{dt}$.
Nevertheless, we can also work with a slower time scale using the small parameter ε such that $\tau := \varepsilon t$.

Definition 2.2. The time scale associated to time t will be called the *fast time scale* and the time scale associated to time τ will be called the *slow time scale*.

Note that the fast-slow system (2.1) can be rewritten in terms of the slow time variable τ. The fast time t is expressed in terms of the slow time τ with the relation $t = \tau/\varepsilon$, so we can write the following:

$$\begin{cases} x' = \frac{dx}{dt} = \frac{dx}{d\left(\frac{\tau}{\varepsilon}\right)} = \varepsilon \frac{dx}{d\tau} = f(x,y,\varepsilon), \\ y' = \frac{dy}{dt} = \frac{dy}{d\left(\frac{\tau}{\varepsilon}\right)} = \varepsilon \frac{dy}{d\tau} = \varepsilon g(x,y,\varepsilon). \end{cases} \tag{2.2}$$

Therefore, we end up with a topologically equivalent system of the form

$$\begin{cases} \varepsilon \dot{x} = f(x,y,\varepsilon), \\ \dot{y} = g(x,y,\varepsilon), \end{cases} \tag{2.3}$$

where we have used the *dot* notation, i.e., $\dot{x} := \frac{dx}{d\tau}$ and $\dot{y} := \frac{dy}{d\tau}$.
Both systems belong to a more general class of problems in *singular perturbation theory* ([15]). It studies systems with a parameter for which the solutions of the problem at a limit value of the parameter are different in nature from the limit of the solutions of the general problem, so the limit is *singular*. This is why the equivalent systems (2.1) and (2.3) may be called *singularly perturbed systems*.

Now, since ε is a considerably small parameter that differentiates the speed of the dynamics of the two types of variables, let us consider the limit case where $\varepsilon \to 0$. Taking this limit in both the fast system (2.1) and the slow system (2.3), we obtain two different but related problems.
First, if we consider the system with respect to the fast time scale (2.1):

Definition 2.3. As $\varepsilon \to 0$ in the system (2.1), we obtain the *fast subsystem*

$$\begin{cases} x' = f(x,y,0), \\ y' = 0, \end{cases} \tag{2.4}$$

which is called the *fast problem* or the *layer problem*. The vector field $X_0(x,y) = (f(x,y,0),0)$ defines the *fast flow*.

In this case, note that the solution for the slow variables contained in vector y is constant and they could be treated as additional parameters for the equations associated to x'. Therefore, since $x \in \mathbb{R}^m$, the layer problem is an m-dimensional problem where we would have to solve an m-dimensional system of differential equations.
Now, if we consider the system with respect to the slow time scale (2.3):

Definition 2.4. As $\varepsilon \to 0$ in the system (2.3), we obtain the *slow subsystem*

$$\begin{cases} 0 = f(x,y,0), \\ \dot{y} = g(x,y,0), \end{cases} \tag{2.5}$$

which is called the *slow problem* or the *reduced problem*. It defines the *slow flow* denoted by X_R.

In this case, note that, since $y \in \mathbb{R}^n$, we have a set of n differential equations subject to the constraints of the algebraic equations $f(x,y,0) = 0$. Therefore, the reduced problem is an n-dimensional differential algebraic problem.

2.2 Equilibria and Manifolds

The constraints $f(x,y,0) = 0$ play an important role in the analysis of fast-slow systems. More generally, the study of the *equilibrium points* (also known as *critical* or *fixed points*) of a dynamical system is always a very informative tool to learn about the equilibrium states of the system and the nature of the solutions. The equilibrium points of a dynamical system modeled by a set of first-order differential equations are the ones where the first derivatives of the dynamical variables vanish simultaneously. Notice that, in the layer problem (2.4), the derivative of the slow variables is always zero, but the derivative of the fast variables vanishes precisely in the points where $f(x,y,0) = 0$. We give a special name to this set.

Definition 2.5. Let $x \in \mathbb{R}^m$, $y \in \mathbb{R}^n$ and let $f : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^m$ be a smooth function. The set

$$C_0 = \{(x,y) \in \mathbb{R}^m \times \mathbb{R}^n \mid f(x,y,0) = 0\} \tag{2.6}$$

is called the *critical manifold* of the layer problem (2.4).

Obviously, since $f(x,y,0) = 0$ introduces m constraints, the dimension of the manifold is $m+n-m = n$ assuming that the jacobian Df evaluated along C_0 has full rank. Moreover, since we assumed that f is a smooth function, the critical manifold is a differentiable manifold. As will be shown later, this set will have a special relevance, since a lot of properties of a singularly perturbed system can be deduced from the properties of the critical manifold.
Having defined the critical manifold that contains all equilibria, we will pay attention to the subsets that contain only *hyperbolic equilibria*.

Definition 2.6. For a general system, a *hyperbolic equilibrium* is a critical point such that the jacobian of the function that describes such system evaluated at that point has no eigenvalues with real part equal to zero. Suppose that all equilibria of a system are hyperbolic:

- If the jacobian only has eigenvalues with negative real part, then the equilibrium point is called an *attractor*.
- If the jacobian only has eigenvalues with positive real part, then the equilibrium point is called a *repellor*.
- If the jacobian has some eigenvalues with positive real part and some eigenvalues with negative real part, then the equilibrium point is called a *saddle point*.

Similar concepts can be extended for subsets of the critical manifold.

Definition 2.7. Let C_0 be the critical manifold of the layer problem and consider a subset $S_0 \subseteq C_0$. If the jacobian with respect to the fast variables $D_x f(x^*, y^*, 0)$ has no eigenvalues with zero real part for all $(x^*, y^*, 0) \in S_0$, then S_0 is called *normally hyperbolic*. Equivalently, S_0 is normally hyperbolic if and only if x^* is a hyperbolic equilibrium of the fast subsystem for $y = y^*$ for every $(x^*, y^*, 0) \in S_0$ (in other words, x^* is a hyperbolic equilibrium of $x' = f(x, y^*, 0)$). Analogously,

- A normally hyperbolic subset $S_0^a \subseteq S_0$ is called *attracting* if every eigenvalue of the jacobian $D_x f(x^*, y^*, 0)$ has negative real part for all $(x^*, y^*, 0) \in S_0^a$.
- A normally hyperbolic subset $S_0^r \subseteq S_0$ is called *repelling* if every eigenvalue of the jacobian $D_x f(x^*, y^*, 0)$ has positive real part for all $(x^*, y^*, 0) \in S_0^r$.
- A normally hyperbolic subset $S_0^s \subseteq S_0$ is called of *saddle type* if it is neither attracting nor repelling.

These definitions have an intuitive importance since the layer problem (2.4) will describe the dynamics of the solutions flowing *away* from S_0^r and the dynamics of the solutions flowing *towards* S_0^a. Furthermore, note that the existence of attracting and repelling manifolds is related to the notion of *stable* and *unstable manifolds*, respectively.
First, consider the linear case:

Definition 2.8. Let $x' = Ax$ be a k-dimensional system of linear differential equations (potentially, the linearized system associated to a non-linear system where $A = Df(p)$ being p a hyperbolic equilibrium point).

- We define the *stable eigenspace* E^s associated to p as the subspace generated by the eigenvectors of A whose corresponding eigenvalues have negative real parts.
- We define the *unstable eigenspace* E^u associated to p as the subspace generated by the eigenvectors of A whose corresponding eigenvalues have positive real parts.

More generally, for the full nonlinear case:

Definition 2.9. Consider a k-dimensional system of autonomous differential equations and let ϕ_t be the flow generated by the system (that is, the evolution map that takes some initial condition as input and gives the state of the variables after time t as an output). Then, for an equilibrium point p:

- We define the *stable manifold* as

$$W^s(p) = \{x \in \mathbb{R}^k \mid \lim_{t \to +\infty} \phi_t(x) = p\}. \tag{2.7}$$

  That is, the set of points such that any solution with one of these points as the initial condition approaches the equilibrium point p as $t \to +\infty$.

- We define the *unstable manifold* as

$$W^u(p) = \{x \in \mathbb{R}^k \mid \lim_{t\to-\infty} \phi_t(x) = p\}. \tag{2.8}$$

  That is, the set of points such that any solution with one of these points as the initial condition approaches the equilibrium point p as $t \to -\infty$.

The analytical computation of these global stable and unstable manifolds is usually not possible and we would resort to numerical methods. However, we have a theorem that guarantees the existence of *local manifolds* around equilibrium points, which could be computed to give an approximation in a small region.

Theorem 2.1. *(Existence of local stable and unstable manifolds).*
Consider a system of differential equations with a hyperbolic equilibrium point p. Let E^s and E^u be the associated stable and unstable eigenspaces. Then, there exist local stable and unstable manifolds $W^s_{loc}(p)$ and $W^u_{loc}(p)$ with the same dimension as E^s and E^u, respectively. Furthermore,

- *(i) $W^s_{loc}(p)$ is tangent to E^s and $W^u_{loc}(p)$ is tangent to E^u at the equilibrium point p. Moreover, $W^s_{loc}(p)$ and $W^u_{loc}(p)$ have the asymptotic properties of E^s and E^u, respectively.*
- *(ii) Both local manifolds are as smooth as the original function that describes the non-linear system.*

When dealing with hyperbolic equilibrium points, this theorem is fundamental to establish a relationship between the phase portraits of the linearized system and the original nonlinear system near an equilibrium point.
Now, we can return to the study of fast-slow systems and extend this notion to build local stable and unstable manifolds of normally hyperbolic manifolds which are subsets of the critical manifold of the layer problem.

Definition 2.10. Let S_0 be a normally hyperbolic manifold contained in the critical manifold C_0. We define the local stable (superscript s) and unstable (superscript u) manifolds of S_0 as

$$W^s_{loc}(S_0) = \bigcup_{p\in S_0} W^s_{loc}(p), \qquad W^u_{loc}(S_0) = \bigcup_{p\in S_0} W^u_{loc}(p). \tag{2.9}$$

In either case, the local manifold associated to a normally hyperbolic manifold is built by joining the local manifolds associated to each individual equilibrium point of the stable/unstable type in the normally hyperbolic manifold.

2.3 Fenichel Theory

The previous section is fundamental to understand the main results of this section concerning the study of singular perturbation problems with normally hyperbolic manifolds from the perspective of a geometric theory. This is called **Fenichel Theory** and it is introduced in [5]. By taking the singular limit $\varepsilon \to 0$, we would like to deduce information for the case when ε is small, but different from zero.
There are multiple general theorems within the theoretical framework. We will now see some statements of the main classical Fenichel Theorem and provide a sketch of the proof.

Theorem 2.2. *(Classical Fenichel Theorem)*
Let $X_\varepsilon(x,y) = (f(x,y,\varepsilon), \varepsilon g(x,y,\varepsilon))$ be a family of vector fields given by a C^r-fast–slow vector field ($r \geq 2$). Let X_R be the slow flow. Let $S_0 \subset C_0$ be an n-dimensional properly embedded C^r-submanifold of $\mathbb{R}^{m+n}$. Suppose that S_0 is normally hyperbolic and has m^s eigenvalues with negative real part and m^u eigenvalues with positive real part. Then, for ε sufficiently small,

- *(i) There exists a family of C^r-manifolds S_ε such that S_ε is a normally hyperbolic locally invariant manifold for X_ε.*
- *(ii) There exist families of $(n+m^s)$- and $(n+m^u)$-dimensional local stable and unstable manifolds $W^s_{loc}(S_\varepsilon)$ and $W^u_{loc}(S_\varepsilon)$.*
- *(iii) All previously mentioned families of manifolds are C^{r-1} in ε.*
- *(iv) The vector field given by*

$$X_{C,\varepsilon}(p) = \begin{cases} \frac{1}{\varepsilon} X_\varepsilon(p) & \text{if } \varepsilon \neq 0 \text{ and } p \in S_\varepsilon, \\ X_R(p) & \text{if } \varepsilon = 0 \text{ and } p \in S_0 \end{cases}$$

is a C^r-vector field on S_ε and $X_{C,\varepsilon}$ is a C^{r-1} family of vector fields in ε.

Proof. (Sketch)
Statements (i) and (ii) follow directly from invariant manifold theorems (see Theorems 2.3.5 and 2.3.6 in [12]).
For (iv), we first have to prove continuity, i.e., we must have

$$X_R(p_\varepsilon) = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} X_\varepsilon(p_\varepsilon) \tag{2.10}$$

for a sequence of points $p_\varepsilon \in S_\varepsilon$.
Note that $\frac{X_\varepsilon}{\varepsilon} = \left(\frac{f}{\varepsilon}, g\right)$. The components are exactly the functions that define the flow on the slow time scale. Taking the limit $\varepsilon \to 0$, we obtain the expression that defines the slow flow X_R.
Now, checking differentiability,

$$\frac{\partial}{\partial \varepsilon}\left(\frac{X_\varepsilon}{\varepsilon}\right) = \left(\frac{\varepsilon \frac{\partial f}{\partial \varepsilon} - f}{\varepsilon^2}, \frac{\partial g}{\partial \varepsilon}\right). \tag{2.11}$$

For $\varepsilon \neq 0$, this defines a vector field

$$\begin{cases} \dot{x} = \frac{\varepsilon \frac{\partial f}{\partial \varepsilon}(x,y,\varepsilon) - f(x,y,\varepsilon)}{\varepsilon^2}, \\ \dot{y} = \frac{\partial g}{\partial \varepsilon}(x,y,\varepsilon). \end{cases} \iff \begin{cases} \varepsilon^2 \dot{x} = \varepsilon \frac{\partial f}{\partial \varepsilon}(x,y,\varepsilon) - f(x,y,\varepsilon), \\ \dot{y} = \frac{\partial g}{\partial \varepsilon}(x,y,\varepsilon). \end{cases} \tag{2.12}$$

Taking the limit $\varepsilon \to 0$, we get

$$\begin{cases} 0 = f(x,y,0), \\ \dot{y} = \frac{\partial g}{\partial \varepsilon}(x,y,0), \end{cases} \tag{2.13}$$

which is just the same as differentiating the equation that defines the slow flow. The same works for higher order derivatives. The upper bound on the order of differentiability in (iv) along with statement (iii) are proved in theorems 1-4 from Fenichel's paper [6].

□

Now, we are going to see the adapted version with direct application to fast-slow systems (applied by Fenichel in [7]). Nevertheless, since Fenichel Theory gives results about the perturbation of manifolds to nearby manifolds, we first define a distance to clarify the meaning of *nearby*.

Definition 2.11. Let A and B be two non-empty sets in $\mathbb{R}^{m+n}$. The *Hausdorff distance* d_H between both sets is defined as

$$d_H(A,B) = \max\left\{ \sup_{a \in A} \inf_{b \in B} ||a-b|| \, , \, \sup_{b \in B} \inf_{a \in A} ||a-b|| \right\}. \tag{2.14}$$

Finally, we present the desired **First Fenichel Theorem** applied to fast-slow systems.

Theorem 2.3. *(First Fenichel Theorem applied to fast-slow systems).*
Let S_0 be a compact normally hyperbolic submanifold of the critical manifold C_0. Let $f : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^m$ and $g : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ be C^r-smooth functions, where $1 \leq r < \infty$. Then, for a sufficiently small $\varepsilon > 0$, the following hold:

- *(i) There exists a locally invariant manifold S_ε (that is, trajectories can enter or leave S_ε only through its boundary) diffeomorphic to S_0.*
- *(ii) As $\varepsilon \to 0$, S_ε has Hausdorff distance $O(\varepsilon)$ from S_0.*
- *(iii) As $\varepsilon \to 0$, the flow on S_ε converges to the slow flow.*
- *(iv) S_ε is C^r-smooth.*

The existence of S_ε and the characterization of its properties in relation to the critical manifold S_0 is fundamental to understand the actual behavior of fast-slow systems when $\varepsilon \neq 0$. We give a name to these manifolds when the perturbation is not negligible.

Definition 2.12. The manifold S_ε described in Theorem 2.3 is called a *slow manifold*.

An important remark is that this slow manifold S_ε is not necessarily unique. However, in regions that remain at a fixed distance from the boundary of S_ε, all manifolds satisfying the properties in Theorem 2.3 lie at a Hausdorff distance $O(\mathrm{e}^{-K/\varepsilon})$ from each other for some $K > 0$ where $K = O(1)$.
Essentially, Theorem 2.3 tells us that we can start our analysis by studying the simpler system in the limit $\varepsilon \to 0$ and obtain the critical manifold and its properties. Then, we consider the full system with the parameter $0 < \varepsilon \ll 1$ that *perturbs* the equations and the theorem guarantees that the topological structure and behavior of the critical manifold still persist. S_ε will be normally hyperbolic like S_0 and it will have the same stability nature (attracting, repelling, or of saddle type) as S_0 with respect to the fast variables. Furthermore, we will find that any compact subset of the manifold perturbs to a slow manifold S_ε and the flow on the slow manifold is very close (order $O(\varepsilon)$) to the original slow flow.
We can even provide a characterization in terms of the local stable and unstable manifolds described in Definition 2.10. This is stated in the **Second Fenichel Theorem**.

Theorem 2.4. *(Second Fenichel Theorem applied to fast-slow systems)*
Let S_0 be a compact normally hyperbolic submanifold of the critical manifold C_0. Let $f : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^m$ and $g : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ be C^r-smooth functions, where $1 \leq r < \infty$. Then, instead of referring to S_0 and S_ε, all the statements of the First Fenichel Theorem also hold for their associated local manifolds. That is,

- *(i) S_0 can be replaced by $W^s_{loc}(S_0)$ or $W^u_{loc}(S_0)$.*
- *(ii) S_ε can be replaced by $W^s_{loc}(S_\varepsilon)$ or $W^u_{loc}(S_\varepsilon)$.*

In other words, the associated local stable and unstable manifolds $W_{loc}(S_0)$ (with s or u superscript, respectively) in the case $\varepsilon \to 0$ persist after considering the real perturbed system with $\varepsilon \neq 0$. Moreover, the topological and analytical properties in the local stable and unstable manifolds $W_{loc}(S_\varepsilon)$ (with s or u superscript, respectively) are preserved.

2.4 *Canards*

The study of the *canard phenomenon* that will be described in the following lines has its origins in the study of the **Van der Pol oscillator**. It is worth briefly elaborating on the details of this model because it provides a specific example of a fast-slow system and because it is the prototypical example of the

appearance of canards. The Van der Pol oscillator is an oscillator with nonlinear damping. The evolution over time of a magnitude x is described by the following second-order differential equation:

$$x'' + \mu(x^2 - 1)x' + x = a, \tag{2.15}$$

where a is a constant forcing term which serves as a control parameter. The parameter μ governs nonlinearity and we will consider the case where it has a large value ($\mu \gg 1$) so that relaxation oscillations appear.
Notice that this equation is written in terms of the prime notation with the derivatives with respect to the fast time t. Since μ has a large value, we can now define a slow time variable $\tau = t/\mu$ and convert this system into the dot notation:

$$\frac{1}{\mu^2}\ddot{x} + (x^2 - 1)\dot{x} + x = a. \tag{2.16}$$

In order to convert it into a system of two first-order differential equations, we introduce a particular change of variables:

$$\begin{cases} y \equiv \frac{1}{\mu^2}\dot{x} + \frac{1}{3}x^3 - x \implies \dot{y} = \frac{1}{\mu^2}\ddot{x} + (x^2 - 1)\dot{x} = a - x, \\ x \equiv x \implies \frac{1}{\mu^2}\dot{x} = y - \frac{1}{3}x^3 + x. \end{cases} \tag{2.17}$$

Since $\mu \gg 1$, we can define the small parameter $\varepsilon := 1/\mu^2$ such that $0 < \varepsilon \ll 1$. Thus, we obtain the following 2D relaxation oscillator:

$$\begin{cases} \varepsilon\dot{x} = y - \frac{1}{3}x^3 + x \equiv f(x, y, \varepsilon), \\ \dot{y} = a - x \equiv g(x, y, \varepsilon), \end{cases} \tag{2.18}$$

with associated Jacobian

$$J(x, y) = \begin{pmatrix} \frac{1}{\varepsilon}\left(-x^2 + 1\right) & \frac{1}{\varepsilon} \\ -1 & 0 \end{pmatrix}. \tag{2.19}$$

System (2.18) has the form of a (1,1)-fast-slow system. Furthermore, using Definition 2.5, we can compute its critical manifold just by considering the set where $f(x, y, 0) = 0$:

$$C_0 = \left\{ (x, y) \in \mathbb{R}^2 \;\middle|\; y = \frac{1}{3}x^3 - x \right\}. \tag{2.20}$$

Notice that the only critical point is contained in this manifold, i.e., $\left(a, \frac{a^3}{3} - a\right) \in C_0$.
Canards appear in fast-slow systems like the one above. More generally, for a general fast-slow system, we have to consider slow manifolds with the properties that Theorem 2.3 describes about S_ε in relation to the critical manifold. In general, S_0 may have an attracting branch S_0^a and a repelling branch S_0^r. Similarly, applying Theorem 2.3, there will exist an attracting slow manifold S_ε^a and a repelling slow manifold S_ε^r with the properties of the theorem.

Definition 2.13. A *canard* is a part of the solution of a fast–slow system that stays within $O(\varepsilon)$ distance to a repelling branch of a slow manifold S_ε^r for a time that is $O(1)$ on the slow time scale.

Specifically,

Definition 2.14. A *maximal canard* is a solution that lies in the intersection of the slow manifolds S_ε^a and S_ε^r.

Essentially, a canard is a solution of a singularly perturbed system where normal hyperbolicity is lost. These points where the critical manifold is not normally hyperbolic will be often called *fold points* (they can also form lines or manifolds of higher dimension for critical manifolds of higher dimensions).

The solution passes through an attracting slow manifold S_ε^a, then it may go near a point where normal hyperbolicity is lost or a bifurcation point of the critical manifold (in the Van der Pol example, $\left(a, \frac{a^3}{3} - a\right)$ is a bifurcation point with $a = 1$), and then traverses (or stays very close to) a repelling slow manifold S_ε^r for some time. They are difficult to detect because of the high sensitivity with respect to the control parameter and their appearance in exponentially small ranges. In the case of the Van der Pol oscillator, we have transitions between a small amplitude limit cycle and a large amplitude relaxation cycle.

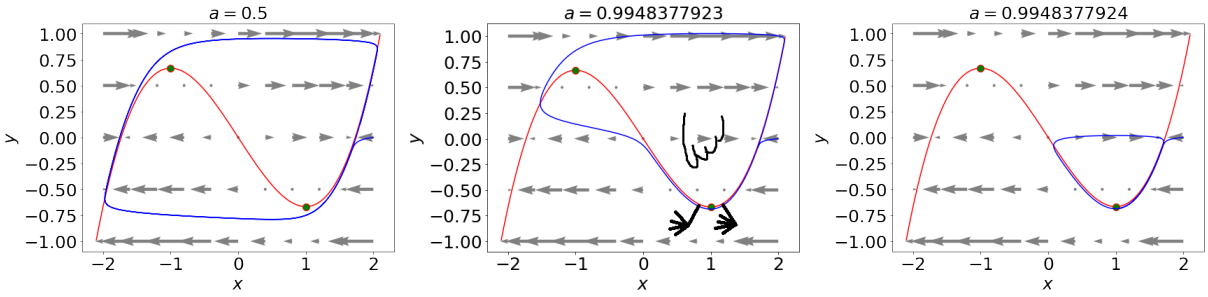


Figure 2.1: Transition (*canard explosion*) through a canard cycle varying the control parameter a. In blue, solutions of the system (2.18) with $\varepsilon = 0.04$ and initial condition $(x(0), y(0)) = (2, 0)$ for different values of a. In red, the critical manifold (2.20). The gray arrows represent the relative magnitude of $\dot{x}$ at each point to reflect that the critical manifold has two outer attracting branches and one inner repelling branch which is followed very closely by a trajectory for some time.

`Canard` is a French word. It has, at least, two meanings in French: `duck` and an informal expression for an `unreliable newspaper`. The interpretation of the first meaning is that the shape of canard cycles in phase portraits somewhat resembles the shape of the animal (as portrayed by the original authors in [4]). The interpretation of the second meaning is that maximal canards follow repelling slow manifolds for some time, which seems counterintuitive at first glance.

As a link to the upcoming chapter, it is worth mentioning that the Van der Pol oscillator lies in the basis of one of the most widely studied excitable cardiomyocyte models: the *FitzHugh-Nagumo Model*. It is a simplified version of the *Hodgkin-Huxley equations*, the first ionic model that represented excitability properties. Its dimensionality was reduced from 4 to 2 equations, but still reflecting the difference between a fast variable and a slow variable. An additional term can also be added in the fast equation to turn it into a system of partial differential equations with diffusion across the spatial dimensions.

Definition 2.15. The FitzHugh-Nagumo PDE Model is defined by the following system of differential equations:

$$\begin{cases} \frac{\mathrm{d}v}{\mathrm{d}t} \equiv v' = \gamma \Delta v + v - \frac{v^3}{3} - w + I, \\ \frac{\mathrm{d}w}{\mathrm{d}t} \equiv w' = \varepsilon \left(v + a - bw\right). \end{cases} \tag{2.21}$$

The dynamical variables are: v, a fast variable representing the voltage in the cell; and w, which is a slow gating variable for the ion channels.
The rest of the terms are: γ, a diffusion coefficient; I, an external current; ε, a small parameter such that $0 < \varepsilon \ll 1$; a, a model parameter to be fitted; and b, a model parameter to be fitted.

The diffusion phenomena in the spatial dimensions can be ignored by setting $\gamma = 0$, so the model would be a system of ODEs. The cubic form of the critical manifold is reminiscent of the Van der Pol oscillator and some dynamical properties are shared.

In the next chapter, we will discuss a couple of slightly more complex models that represent adequately some qualitative characteristics of cardiac cells.

Chapter 3

Application to Cardiomyocyte Models

It is known that cardiac excitability depends on the flow of **ionic currents** across the cell membranes of the cardiomyocytes, as well as on the variable electrical potential. Usually, we will deal with the flow of some ions among Ca^{2+} (*calcium*), Na^{+} (*sodium*) or K^{+} (*potassium*). Therefore, we will study how these currents affect **voltage** inside the cells with respect to the outside. Short changes in voltage are called **action potentials**. They begin with a rise in voltage and end with a fall to the resting level until the process starts again.
Many models with different degrees of complexity have been proposed. In this chapter, we will analyze two models for the electrical activity in the heart: the Karma Model and the Reduced Luo-Rudy Model. These models consist of sets of differential equations which attempt to model the physical phenomena undergone by cardiomyocytes. We describe the main features that characterize these models and the motivation behind the functional form of the equations. Then, we will perform a series of numerical simulations to illustrate the main information that can be extracted. Other interesting results from the perspective of the study of dynamical systems will be shown, including applications of concepts from the previous chapter. Namely, relevant equilibria, orbits, manifolds or canards and their physical interpretation.

3.1 The Karma Model

The **Karma Model** is a reduced version of the *Noble Model*, but with a fast-slow structure as the FitzHugh-Nagumo Model that was introduced before. The Karma Model shows the evolution of the voltage in a cardiac cell after it is stimulated above a certain threshold. In 1994, Karma proposed a general form.

3.1.1 Definition of the Karma Model

Definition 3.1. The general dimensionless Karma PDE Model (with independent fast time scale variable t) is defined by the following system of differential equations:

$$\begin{cases} \frac{\mathrm{d}E}{\mathrm{d}t} \equiv E' = \gamma \Delta E + \frac{1}{\tau_E} f(E,n), \\ \frac{\mathrm{d}n}{\mathrm{d}t} \equiv n' = \frac{1}{\tau_n} g(E,n), \end{cases} \tag{3.1}$$

where

$$\begin{cases} f(E,n) = -E + (E^* - D(n))\, h(E), \\ g(E,n) = R(n)\theta(E - E_n) - (1 - \theta(E - E_n))\, n. \end{cases} \tag{3.2}$$

The dynamical variables are: E, a fast variable representing the electrical **transmembrane potential** in the cell; and n, which is a slow **gating variable** controlling the activation of the ion channels.
The rest of the terms are: γ, a diffusion coefficient; E^*, a parameter to be fitted; τ_E, a time parameter defining the scale for the reaction terms at which E evolves; τ_n, a time parameter defining the scale for the reaction terms at which n evolves; $h(E)$, the *reaction* function; $R(n)$, the *restitution* function; and $D(n)$, the *dispersion* function. Additionally, $\theta(E - E_n)$ is a suitable piecewise function representing that the ion channels remain totally closed below a certain threshold potential (we will fix $E_n = 1$) and start to open when that voltage is surpassed. An adequate option could be the Heaviside step function, i.e., $\theta(E-1) = 0$ for $E \leq 1$ and $\theta(E-1) = 1$ for $E > 1$. We could also consider an extension of that function to make it continuous at the threshold and represent a more slow and realistic transition. Nevertheless, in this case, we will work with a ReLU function, i.e., $\theta(E-1) = \max(0, E-1)$, which is simple and represents correctly the behavior of the ion channels in the desired range.

When at rest, heart muscle cells are polarized, i.e., ion concentrations are different inside and outside the cell membrane carrying different net charge densities. In that case, the potential inside of the cells is negative with respect to the potential outside of the cells. The depolarization occurs when the transmembrane voltage suddenly increases and the ionic currents with charged particles flow across the cell membranes. After the initial spike, the voltage may experience a gentle drop with small oscillations until it finally reaches the final sharp drop that corresponds to the repolarization phase.

Proposition 3.1. *Let $\gamma = 0$ so system (3.2) is a system of ODEs. Assume that the relevant approximate interval for the potential E with a typical choice of parameters is $[0,4]$. A suitable reaction function $h(E)$ in that interval that guarantees the existence of exactly two stable equilibria in the first equation for a fixed value of n corresponding to the polarized and the depolarized states of a cardiomyocyte is a non-negative function with a single maximum such as*

$$h(E) = 2\left(E^2 - \frac{1}{4}E^3\right). \tag{3.3}$$

Proof. The physical intuition for the choice of $h(E)$ is that there must be exactly two stable equilibria for a fixed value of n corresponding to the polarized and depolarized states of a cardiomyocyte. We look at the first equation of the system (3.2), i.e., $-E + (E^* - D(n))h(E)$. We make $h(0) = 0$ so there is an equilibrium at $E = 0$. The derivative with respect to E of such first equation is $-1 + (E^* - D(n))h'(E)$, so stability for a fixed n is guaranteed if we also make $h'(0) = 0$ because $-1 < 0$. Choosing h such that it is positive for $E > 0$ in the desired range, the other stable state can be obtained at a unique maximum of h where the derivative also vanishes and $-1 < 0$. A classical choice that works well in this interval depends on the hyperbolic tangent function with a threshold E_h. However, a more convenient choice is a third degree polynomial interpolating the original function at $E = 0$ and $E = E_h = 3$ and with the same first derivatives at both points:

$$h(E) = (1 - \tanh(E - E_h))\frac{E^2}{2} \approx 2\left(E^2 - \frac{1}{4}E^3\right). \tag{3.4}$$

□

A *restitution curve* $A(d)$ describes the action potential duration as a function of the diastolic interval, i.e., the time that the heart takes to relax after a contraction and refills with blood from the end of an action potential to the beginning of the next one. The *restitution function* $R(n)$ is related to such curve.

Proposition 3.2. *Let r be a control parameter related to the restitution properties of the heart tissue. Consider a simple linear restitution function of the form*

$$R(n) = \frac{1 - (1 - e^{-r})n}{1 - e^{-r}}. \tag{3.5}$$

This leads to the suitable restitution curve

$$A(d) = \tau_n r + \tau_n \ln\left(1 - \left(1 - e^{-r}\right)e^{-\frac{d}{\tau_n}}\right). \tag{3.6}$$

Proof. Consider a general expression for the restitution function and our particular choice:

$$R(n) = \frac{n}{\frac{\mathrm{d}A}{\mathrm{d}d}\big|_{d=\tau_n \ln(1/n)}} = \frac{1 - (1 - \mathrm{e}^{-r})n}{1 - \mathrm{e}^{-r}}. \tag{3.7}$$

Before integrating $\frac{\mathrm{d}A}{\mathrm{d}d}$, consider the inverse evaluation $n = \mathrm{e}^{-\frac{d}{\tau_n}}$, which we substitute:

$$\frac{\mathrm{d}A}{\mathrm{d}d} = \frac{(1 - \mathrm{e}^{-r})\mathrm{e}^{-\frac{d}{\tau_n}}}{1 - (1 - \mathrm{e}^{-r})\mathrm{e}^{-\frac{d}{\tau_n}}} \implies A(d) = \int \frac{(1 - \mathrm{e}^{-r})\mathrm{e}^{-\frac{d}{\tau_n}}}{1 - (1 - \mathrm{e}^{-r})\mathrm{e}^{-\frac{d}{\tau_n}}}\mathrm{d}d = \tau_n \ln\left(1 - (1 - \mathrm{e}^{-r})\mathrm{e}^{-\frac{d}{\tau_n}}\right) + C. \tag{3.8}$$

If a diastolic interval is nonexistent (dead cell), the following action potential is also nonexistent, so the condition $A(0) = 0$ leads to a monotonically increasing curve $A(d)$ with $C = \tau_n r$.

□

Related to this control parameter, we will later introduce the magnitude $n_B := 1 - e^{-r}$. Also, notice that an increase in r implies a decrease in $R(n)$, which in turn implies that the slope of the restitution curve is considerably higher for short diastolic intervals. Biologically, this makes the tissue more susceptible to *alternans*, i.e., oscillatory pulse instabilities that cause the action potential duration to alternate in time.
The dispersion function $D(n)$ relates the dispersion velocity of a pulse and the length of the previous diastolic interval. A simple polynomial relation with a control parameter M will suffice:

$$D(n) = n^M, \qquad n \in (0,1),\ M \gg 1. \tag{3.9}$$

The physical interpretation of the parameter M is better understood by looking at the full partial differential equations model with a non-zero diffusion coefficient γ. Although only the ODE model will be analyzed in this thesis, the analysis of the full PDE model would lead to the study of wave solutions. In particular, traveling pulses. The parameter $M \gg 1$ controls how insensitive the wave-front velocity is with respect to the slow gating variable n.
Substituting all the terms, we get the functions defined in (3.2):

$$\begin{cases} f(E,n) = -E + 2\left(E^* - n^M\right)\left(E^2 - \frac{1}{4}E^3\right), \\ g(E,n) = \frac{1-(1-e^{-r})n}{1-e^{-r}}\theta(E-1) - (1-\theta(E-1))n = \frac{1}{1-e^{-r}}\theta(E-1) - n, \end{cases} \tag{3.10}$$

leading to the following system:

$$\begin{cases} E' = \gamma\Delta E + \frac{1}{\tau_E}\left(-E + 2(E^* - n^M)(E^2 - \frac{1}{4}E^3)\right), \\ n' = \frac{1}{\tau_n}\left(\frac{1}{1-e^{-r}}\theta(E-1) - n\right). \end{cases} \tag{3.11}$$

3.1.2 Fast-Slow Formulation of the Karma Model

We will now proceed to see explicitly the fast-slow nature of the model.

Proposition 3.3. *(Fast-Slow formulation of the Karma Model).*
The system described in (3.11) with $\gamma = 0$ (ODEs) admits a representation as a fast-slow system.

Proof. We are going to try to get a representation as a (1,1)-fast-slow system assuming that the fast variable is E and the slow variable is n. Also, consider $n_B := 1 - e^{-r}$.
In the model, the parameters τ_E and τ_n have the role of adjusting time scales. Since we assume that the equation for n is the slow one and it has the factor $1/\tau_n$, we will assume that τ_n is large since n' must be small. For the fast E equation, τ_E cannot be that large, so we will assume that $\tau_E \ll \tau_n$ and we define $\varepsilon = \tau_E/\tau_n$ such that $0 < \varepsilon \ll 1$.
Without loss of generality, since the model is dimensionless, we will take $\tau_E = 1$ (and, therefore, $\varepsilon = 1/\tau_n$). This is enough to express system (3.11) as a fast-slow system with derivatives with respect to the fast time scale variable t:

$$\begin{cases} E' = -E + 2\left(E^* - n^M\right)\left(E^2 - \frac{1}{4}E^3\right), \\ n' = \varepsilon\left(\frac{1}{n_B}\theta(E-1) - n\right). \end{cases} \tag{3.12}$$

□

For the following analysis, we will mainly use the model in the form of the expression (3.12).

3.1.3 Analysis of the Ordinary Differential Equations Model

We will study the dynamical properties of the Karma Model without considering the diffusion mechanisms across the spatial dimensions. That is, we set $\gamma = 0$ and we study the resulting system of ordinary differential equations.

Proposition 3.4. *The critical manifold of system (3.12) is*

$$C_0 = \left\{ (E,n) \in \mathbb{R}^2 \;\middle|\; E = 0 \quad or \quad n = \sqrt[M]{E^* - \frac{1}{2E\left(1-\frac{1}{4}E\right)}} \right\}. \tag{3.13}$$

The point $(E,n) = (2,1)$ is the only fold point (where the critical manifold is not normally hyperbolic).

Proof. The critical manifold is the set of points where the fast equation (first equation in system (3.12)) vanishes:

$$-E + 2\left(E^* - n^M\right)\left(E^2 - \frac{1}{4}E^3\right) = E\left(-1 + 2\left(E^* - n^M\right)\left(E - \frac{1}{4}E^2\right)\right) = 0. \tag{3.14}$$

The first factor gives $E = 0$ and the second factor gives the second condition solving for n.
Furthermore, by definition of normal hyperbolicity, fold points are computed by considering the jacobian of the function defining the fast equations with respect to the fast variables. In this case, this is the first derivative of E' with respect to E. We see where it vanishes, i.e., the jacobian has a zero eigenvalue.

$$\frac{\partial f}{\partial E} = -1 + 2\left(E^* - n^M\right)\left(2E - \frac{3}{4}E^2\right) = 0. \tag{3.15}$$

Since the fold points are contained in the critical manifold, we introduce the conditions that define C_0. Clearly, $E = 0$ leads to an impossible equality in the above equation, so we substitute the other branch that gives n^M:

$$\frac{\partial f}{\partial E} = -1 + 2 \cdot \frac{1}{2E\left(1-\frac{1}{4}E\right)} \cdot E\left(2 - \frac{3}{4}E\right) = 0. \tag{3.16}$$

The solution is $E = 2$. Substituting in the equation for n in (3.13), this gives $n = 1$.

□

From now on, let us fix $E^* = 1.5$, $\varepsilon = 0.01$, $M = 4$ and $n_B = 0.5$.

Proposition 3.5. *The Karma ODE Model (i.e., (3.12) with $\gamma = 0$) has 3 equilibrium points:*

- *The origin is a stable fixed point.*
- *Further in the line $n = 0$, the intersection with the critical manifold is a saddle point.*
- *The intersection of the line $n = 2(E-1)$ with the critical manifold is an unstable fixed point.*

Proof. The equilibria of the system are the points where both equations vanish.
Since the piecewise function $\theta(E-1) = \max(0, E-1)$ has two pieces, we have to analyze both cases separately.
First, assume $E < 1$ such that $\theta(E-1) = 0$. In that case, the second equation reads

$$n' = -0.01n, \tag{3.17}$$

which vanishes if and only if $n = 0$. Substituting in the first equation, it reads

$$E' = -E + 3\left(E^2 - \frac{1}{4}E^3\right) = E\left(-1 + 3\left(E - \frac{1}{4}E^2\right)\right). \tag{3.18}$$

There are three solutions for E, which are $E = 0$ and $E = 2 \pm \frac{2}{3}\sqrt{6}$. We discard the solution with $E > 1$ since we assumed $E < 1$ in order to work with a specific piece of θ. This leaves the equilibria $(E,n) = (0,0)$ and $(E,n) \approx (0.367,0)$.
Now, assume $E > 1$ such that $\theta(E-1) = E-1$. In that case, the second equation reads

$$n' = 0.01\,(2(E-1) - n)\,. \tag{3.19}$$

Clearly, it vanishes with $n = 2(E-1)$ which we must substitute in the first equation. In this case, $E \neq 0$, so the first equation will vanish if and only if

$$-1 + \left(3 - 32(E-1)^4\right)\left(E - \frac{1}{4}E^2\right) = 0, \tag{3.20}$$

There are two solutions such that $E < 1$ and one solution that leads to $n \gg 1$. The remaining root is the only valid solution, $E \approx 1.496$ and it gives the remaining equilibrium point: $(E,n) \approx (1.496, 0.992)$.
The jacobian of system (3.12) is

$$J(E,n) = \begin{pmatrix} -1 + \left(3 - 2n^4\right)\left(2E - \frac{3}{4}E^2\right) & -8n^3\left(E^2 - \frac{1}{4}E^3\right) \\ 0.02\theta'(E-1) & -0.01 \end{pmatrix}. \tag{3.21}$$

Evaluating at the three equilibria, we obtain

$$J(0,0) = \begin{pmatrix} -1 & 0 \\ 0 & -0.01 \end{pmatrix}, \quad J(0.367,0) \approx \begin{pmatrix} 0.899 & 0 \\ 0 & -0.01 \end{pmatrix}, \quad J(1.496,0.992) \approx \begin{pmatrix} 0.397 & -10.941 \\ 0.02 & -0.01 \end{pmatrix}. \tag{3.22}$$

The first jacobian has two negative eigenvalues. Therefore, the first point is a stable point (node).
The second jacobian has a positive eigenvalue and a negative eigenvalue. Therefore, the second point is a saddle point. Since the matrix is diagonal, we readily see that the associated unstable eigenspace goes in the horizontal direction (eigenvector $(1,0)$) near the point in the (E,n)-plane and the associated stable eigenspace goes in the vertical direction (eigenvector $(0,1)$) near the point in the (E,n)-plane.
The third jacobian has two complex eigenvalues with positive real part. Therefore, the third point is an unstable point (a focus, although the strength of the critical manifold masks the spiraling).

□

A numerical simulation of a typical trajectory will help to visualize all the elements computed above. We will solve the system numerically for 600 units of fast time and observe the phase portrait and the corresponding time series of E and n (for these, only the first 60 units of time are shown to visualize correctly the pulse).

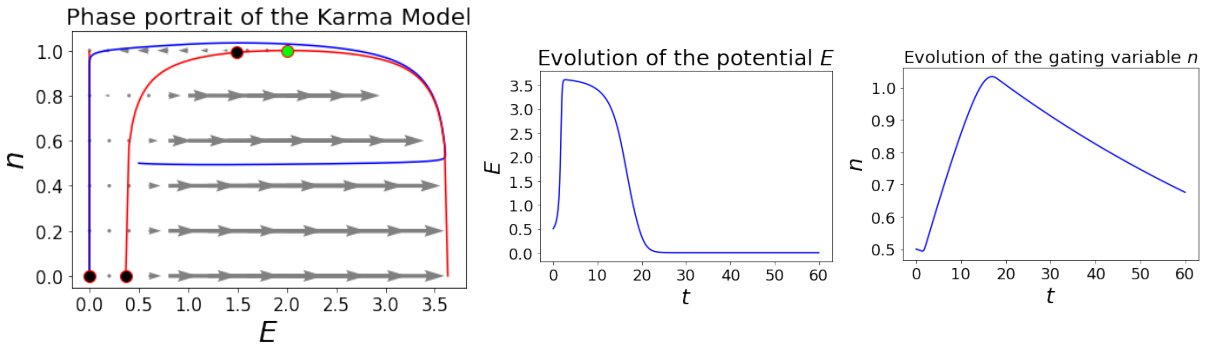


Figure 3.1: On the left, a phase portrait of the system (3.12). In blue, a solution of the system during 600 units of time with initial condition $(E(0),n(0)) = (0.5,0.5)$. In red, the critical manifold computed in (3.13). In green, the unique fold point. In black, the three equilibria. The gray arrows represent the relative magnitude of E' in several points of the (E,n)-plane. On the center and right, time series describing the time evolution of the two dynamical variables.

We see the two disconnected parts of the critical manifold. The vertical line on the left is an attracting branch of the critical manifold. The part on the right has a repelling branch on the left and an attracting branch on the right. They are separated by the unique fold point. Also, in the time series, we see the pulse of the action potential with a quick depolarization and a more gentle drop during the repolarization. After the time shown, the gating variable would continue to decay exponentially to zero. We also see how the electric pulse is almost instantaneous compared to the speed of opening and closing of the ion channels, reflecting the importance of considering different time scales in some biological systems.

In the singular limit $\varepsilon \to 0$, we construct the *candidate orbits*. If the initial point is away from the critical manifold, the orbit must follow the fast flow. The orbits are repelled from the central repelling branch. If the initial point is to the left of such branch, the trajectory will fall to the leftmost attracting branch and will follow the slow flow towards the stable equilibrium $(0,0)$. If the initial point is to the right of the repelling branch, the trajectory will fall to the rightmost attracting branch. Then, the orbit follows the slow flow through the attracting branch up to the fold point. Then, there is a fast flow jump towards the other attracting branch. Finally, the trajectory follows the slow flow and descends to the stable origin through this attracting branch. If the initial point is in an attracting branch of the critical manifold, it will follow one of the dynamics explained above until the stable equilibrium point. If the initial point is in the unstable branch and it is not the unstable equilibrium or the saddle (there would be no dynamics), the orbit may go to the fold point and jump to the left branch (starting above the unstable point) or will descend towards the saddle (starting below the unstable point).
Furthermore, we recall that the unstable equilibrium point and the saddle point are in the same branch of the critical manifold, with the latter having its stable manifold tangent to the critical manifold at the saddle point. In fact, this is related to the fact that there exists a heteroclinic connection between the saddle and the unstable point (a solution connecting them). Thus, we claim that, at least in the limit $\varepsilon \neq 0$, every orbit will eventually converge to the stable equilibrium at the origin or it is one of the other fixed points or it is the heteroclinic connection joining them. We will now proceed to see that the objects are preserved when $\varepsilon > 0$.

Proposition 3.6. *Away from the heteroclinic connection of the critical manifold between the saddle point and the unstable point, the candidate orbits, which may be found in the singular limit $\varepsilon \to 0$, can be perturbed to solution curves of the full system with $0 < \varepsilon \ll 1$.*

Proof. These are consequences of the Fenichel Theory, whose main results were presented in the previous chapter specifically for their application in fast-slow systems. However, a small workaround is needed because Fenichel's theorems require differentiability. We are dealing with a system with the function $\theta(E-1) = \max(0, E-1)$, which is not differentiable at $E = 1$.
The idea is that the results of Fenichel Theory are local around compact subsets of the critical manifold that we consider. For this reason, we can extend the systems on each side to the entire real plane and the theorems are still applicable. In summary, one could consider the system with $\theta(E-1) = 0$ or the system with $\theta(E-1) = E-1$:

$$\begin{cases} E' = -E + 2\left(E^* - n^M\right)\left(E^2 - \frac{1}{4}E^3\right), \\ n' = -\varepsilon n. \end{cases} \qquad \begin{cases} E' = -E + 2\left(E^* - n^M\right)\left(E^2 - \frac{1}{4}E^3\right), \\ n' = \varepsilon\left(\frac{1}{n_B}(E-1) - n\right). \end{cases} \tag{3.23}$$

with $(E,n) \in \mathbb{R}^2$ in both cases and apply the results of Fenichel Theory. The only non-normally hyperbolic point of the critical manifold is the fold point $(E,n) = (2,1)$. Therefore, away from this point, we can take any compact subset and apply the First Fenichel Theorem to perturb the attracting and repelling branches to locally invariant manifolds of the full system.

□

Before proceeding to the verification of the existence of the heteroclinic connection in the actual system, we must first state an important theorem in dynamical systems: the Poincaré-Bendixson Theorem.

Definition 3.2. Consider a differentiable real dynamical system with flow ϕ_t. A point x is an ω-limit (respectively, α-limit) of a point x_0 if there exists a sequence $\{t_n\}_{n\in\mathbb{N}}$ such that

$$\lim_{n\to\infty} t_n = \infty \text{ (respectively, } -\infty) \qquad \text{and} \qquad \lim_{n\to\infty} \phi_{t_n}(x_0) = x. \tag{3.24}$$

For an orbit Γ, we say that x is an ω-limit (α-limit) point of Γ if it is an ω-limit (α-limit) point of some point in the orbit. The set of all ω-limit (α-limit) points for an orbit Γ is called an ω-limit (α-limit) set for Γ.

Theorem 3.1. *(General Poincaré-Bendixson Theorem)*
Consider a differentiable dynamical system defined in an open subset of $\mathbb{R}^2$. Let M be a positively (respectively, negatively) invariant region. That is, a set where orbits through any point $p \in M$ stay trapped for any amount of forward time (respectively, backward time). Suppose that M contains a finite number of fixed points. Then, the ω-limit (respectively, the α-limit) set of an orbit Γ passing through p is either a fixed point, a periodic orbit or a set with a finite number of fixed points connected by homoclinic and heteroclinic orbits.

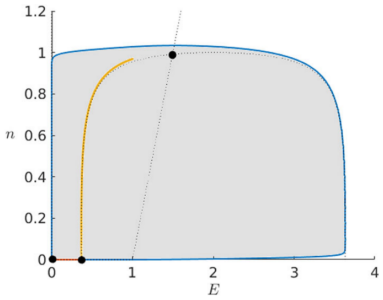


Figure 3.2: Invariant set of the Karma Model bounded by the blue and orange unstable manifolds associated to the saddle equilibrium. In yellow, its stable manifold (extracted from [9]).

Proposition 3.7. *The heteroclinic connection of the critical manifold in the singular limit $\varepsilon \to 0$ can also be perturbed to a heteroclinic orbit between the unstable equilibrium and the saddle equilibrium with $0 < \varepsilon \ll 1$.*

Proof. In the actual system with $0 < \varepsilon \ll 1$, we notice that we can build an invariant set. The unstable manifolds of the saddle (locally tangent to unstable eigenspaces by Theorem 2.1) are computed by extending the flow away the saddle point to the left and to the right, since the horizontal direction is given by the eigenvector $(1,0)$ associated to the positive eigenvalue of the jacobian evaluated at the saddle point (see Proposition 3.5). By the analysis of candidate orbits above, we know that these unstable manifolds of the saddle point will converge to the stable origin, thus creating a trapping region.
We checked that we have an invariant region with a finite number of isolated fixed points. By the analysis of candidate orbits above, we deduce that every orbit starting away from the heteroclinic connection will eventually converge to the stable origin. Therefore, there are not any periodic orbits or limit cycles in this invariant set. We will apply Poincaré-Bendixson Theorem for backward time (for α-limit sets). When time goes in the opposite direction, the unstable manifolds of the saddle point that let orbits flow away, now they do not. The orbits can flow away in the direction of the formerly stable manifold (locally indicated by the direction of the eigenvector $(0,1)$; see Proposition 3.5), now unstable in backward time. We consider the Poincaré-Bendixson Theorem for the α-limit set of the orbit that flows along the formerly stable manifold of the saddle. Clearly, the α-limit set cannot be just a single fixed point, there are no periodic orbits and the existence of a homoclinic orbit (connecting the saddle with itself) would imply the existence of periodic orbits inside. Since the formerly unstable equilibrium point is now stable in backward time, the connection must be the heteroclinic orbit. □

3.2 The Reduced Luo–Rudy Model

In this section, we will analyze the reduced **Luo–Rudy** mammalian ventricular cell model. In 1991, Luo and Rudy introduced a model to explain the electrical behavior of a mammalian ventricular cardiomyocyte based on experimental data available at that time. However, a popular reduction of the model was later proposed to investigate the mechanisms of **early afterdepolarizations** (EADs) when decreasing the stimulation frequency. Afterdepolarizations are abnormal depolarizations of cardiomyocytes that interrupt some phase of the cardiac action potential conduction process. They can result in a number of pathologies such as *torsades de pointes*, tachycardia, and other arrhythmias.

3.2.1 Definition of the Reduced Luo-Rudy Model

The rate of change of the transmembrane potential V can be related to the membrane capacitance C_m (we will work with this magnitude per 1 cm^2 of membrane area) and the total current I with $\frac{\mathrm{d}V}{\mathrm{d}t} = -\frac{I}{C_m}$. The total current is $I = I_{Ca} + I_K + I_{sti}$, where I_{sti} is an external stimulus current. In this case, we will mainly work with $I_{sti} = 0$. The functional form of the Ca and K currents is based on the Ohm's law. A ionic current is equal to the transmembrane voltage V with respect to a reference value E_{Ca} or E_K and multiplied by a conductance G_{Ca} or G_K (inverse of resistance). Additionally, as usual, the activation and inactivation of ion channels is modeled by gating variables. For the Ca current, we consider an activation gating variable d approximated by its steady-state function $d_\infty(V)$, assuming that it asymptotically reaches a time-independent value that we can use to approximate its full behavior and an inactivation gating variable f. For the K current, we consider a time-dependent activation gating variable x and a time-independent inactivation gating variable x_i that we will set to 1 for simplicity. This leaves three differential equations for V, f and x. There will also be characteristic time constants τ_V, τ_f and τ_x.

Definition 3.3. The Reduced three-variable Luo-Rudy Model (with $I_{sti} = 0$) is defined by the following system of differential equations:

$$\begin{cases} C_m \frac{\mathrm{d}V}{\mathrm{d}t} = -(I_{Ca} + I_K) \equiv h(V,f,x), \\ \frac{\mathrm{d}f}{\mathrm{d}t} = \frac{f_\infty(V) - f}{\tau_f} \equiv g_1(V,f,x), \\ \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{x_\infty(V) - x}{\tau_x} \equiv g_2(V,f,x). \end{cases} \tag{3.25}$$

The dynamical variables are V, the **transmembrane potential**; f, the **inactivation gating variable** for the calcium current I_{Ca}; and x, the **activation gating variable** for the potassium current I_K.
The currents are defined as follows:

$$\begin{cases} I_{Ca} = G_{Ca} \cdot d_\infty(V) \cdot f \cdot (V - E_{Ca}), \\ I_K = G_K \cdot x \cdot (V - E_K), \end{cases} \tag{3.26}$$

The steady states can be modeled with the following steady-state functions that take values between 0 and 1:

$$f_\infty(V) = \frac{1}{\mathrm{e}^{\frac{V - E_f}{\theta_f}} + 1}, \qquad x_\infty(V) = \frac{1}{\mathrm{e}^{\frac{E_x - V}{\theta_x}} + 1}, \qquad d_\infty(V) = \frac{1}{\mathrm{e}^{\frac{E_d - V}{\theta_d}} + 1}, \tag{3.27}$$

where E_f, θ_f, E_x, θ_x, E_d, θ_d are model parameters.

A somewhat realistic approximation of the parameters of the model could be the following values that will be used for later analysis:

C_m (μF/cm^2)	0.5
τ_f (ms)	80
τ_x (ms)	300

G_{Ca} (mS/cm^2)	0.025
E_{Ca} (mV)	100
G_K (mS/cm^2)	variable
E_K (mV)	-80

E_f (mV)	-20
θ_f (mV)	8.60
E_x (mV)	-40
θ_x (mV)	5.00
E_d (mV)	-35
θ_d (mV)	6.24

Table 3.1: Numerical values for the parameters of the Reduced Luo-Rudy Model.

3.2.2 Fast-Slow Formulation of the Reduced Luo-Rudy Model

The Reduced Luo-Rudy Model can be transformed into the standard form of a fast-slow system with the appropriate assumptions in order to apply the theory discussed in the previous chapter.

Proposition 3.8. *(Fast-Slow formulation of the Reduced Luo-Rudy Model).*
The system of differential equations described in (3.25) admits a representation as a fast-slow system.

Proof. First, we need to decide which equations will be fast and which ones will be slow. To do this, we have to pay attention to the actual numerical values. In table (3.1), we showed fairly realistic values (supported by physical laws or experimental data) for the time constants τ_f and τ_x. We are using $\tau_f =$ 80 ms and $\tau_x = 300$ ms. The total conduction of electricity through the Ca and K channels can be seen as an electrical current subject to the resistance of two resistors in parallel. In that case, the equivalent total resistance is $1/(G_{Ca}+G_K)$, being G_{Ca} and G_K the conductances (per unit area), i.e., the inverses of the resistances of each individual ionic channel. The lowest conductance G_K that we will examine is $G_K = 0.03345$ mS/cm^2. The time constant for V can be estimated as the capacitance multiplied by the resistance (inverse of total conductance):

$$\tau_V = \frac{C_m}{G_{Ca}+G_K} = \frac{0.5\ \mu\text{F/cm}^2}{0.025\ \text{mS/cm}^2 + 0.03345\ \text{mS/cm}^2} \approx 8.6\ \text{ms}. \tag{3.28}$$

We see that $\tau_V < \tau_f < \tau_x$ being τ_V considerably lower than the others. We could consider V as a fast variable and x as a slow variable. On the other hand, f could be considered fast or slow, and different independent analyses could be proposed depending on the choice. In this case, we will take f as another slow variable, so we are seeking a (1,2)-fast-slow system.
First, notice that we can rescale system (3.25) and obtain a new fast time scale variable $\tilde{t} = t/C_m$. Then, we can define the parameter $0 < \varepsilon \ll 1$ as C_m/τ_f or C_m/τ_x indifferently (both quotients are small enough for our purposes and we will assign them the same parameter ε). Therefore, we end up with the following pair of fast time scale ($\tilde{t}$) and slow time scale ($\tau = \varepsilon\tilde{t}$) systems:

$$\begin{cases} \frac{\mathrm{d}V}{\mathrm{d}\tilde{t}} = -(I_{Ca}+I_K), \\ \frac{\mathrm{d}f}{\mathrm{d}\tilde{t}} = \varepsilon(f_\infty(V)-f), \\ \frac{\mathrm{d}x}{\mathrm{d}\tilde{t}} = \varepsilon(x_\infty(V)-x). \end{cases} \iff \begin{cases} \varepsilon\frac{\mathrm{d}V}{\mathrm{d}\tau} = -(I_{Ca}+I_K), \\ \frac{\mathrm{d}f}{\mathrm{d}\tau} = f_\infty(V)-f, \\ \frac{\mathrm{d}x}{\mathrm{d}\tau} = x_\infty(V)-x. \end{cases} \tag{3.29}$$

□

Proposition 3.9. *The critical manifold associated to the fast-slow system is*

$$C_0 = \left\{ (V,f,x) \in \mathbb{R}^3 \;\middle|\; f = -\frac{G_K \cdot x \cdot (V-E_K)}{G_{Ca} \cdot d_\infty(V) \cdot (V-E_{Ca})} \right\}. \tag{3.30}$$

Proof. The manifold is defined by the points where the first equation in the systems of (3.29) vanishes.

$$h(V,f,x) = -(I_{Ca}+I_K) = -[G_{Ca} \cdot d_\infty(V) \cdot f \cdot (V-E_{Ca}) + G_K \cdot x \cdot (V-E_K)] = 0 \implies \tag{3.31}$$

$$\implies f = -\frac{G_K \cdot x \cdot (V - E_K)}{G_{Ca} \cdot d_\infty(V) \cdot (V - E_{Ca})}. \tag{3.32}$$

□

The critical manifold that was obtained in Proposition 3.9 is a smooth surface in $\mathbb{R}^3$ (we could even express $f = f(V,x)$ analytically) and its different sheets will be separated by curves on the surface. These are the *fold lines*, i.e., fold bifurcations of the layer problem.

Proposition 3.10. *The fold lines $F^\pm$ separating the different sheets of the critical surface are given by*

$$F^\pm = \left\{ (V,f,x) \in \mathbb{R}^3 \middle| f = -\frac{G_K x(V-E_K)}{G_{Ca} d_\infty (V-E_{Ca})} = -\frac{G_K x}{G_{Ca}\left(\frac{dd_\infty}{dV}(V-E_{Ca}) + d_\infty\right)} \neq 0 \quad \textit{and} \quad \frac{d^2 d_\infty}{dV^2}(V-E_{Ca}) + 2\frac{dd_\infty}{dV} \neq 0 \right\}. \tag{3.33}$$

Proof. We look for the points in the critical manifold where the partial derivative of the function $h(V,f,x)$ defining the evolution of the fast variable V with respect to that variable vanishes and the second partial derivative does not (nondegeneracy):

$$F^\pm = \left\{ (V,f,x) \in C_0 \middle| \frac{\partial h}{\partial V} = 0, \frac{\partial^2 h}{\partial V^2} \neq 0 \right\}. \tag{3.34}$$

Recall that $h(V,f,x) = -(I_{Ca} + I_K) = -[G_{Ca} d_\infty(V) f (V - E_{Ca}) + G_K x (V - E_K)]$. Therefore,

$$\frac{\partial h}{\partial V} = -\left[G_{Ca} f \left(\frac{\mathrm{d}d_\infty}{\mathrm{d}V}(V - E_{Ca}) + d_\infty \right) + G_K x \right], \tag{3.35}$$

$$\frac{\partial^2 h}{\partial V^2} = -G_{Ca} f \left(\frac{\mathrm{d}^2 d_\infty}{\mathrm{d}V^2}(V - E_{Ca}) + 2\frac{\mathrm{d}d_\infty}{\mathrm{d}V} \right), \tag{3.36}$$

where

$$d_\infty = \frac{1}{\mathrm{e}^{\frac{E_d - V}{\theta_d}} + 1}, \qquad \frac{\mathrm{d}d_\infty}{\mathrm{d}V} = \frac{\mathrm{e}^{\frac{E_d - V}{\theta_d}}}{\theta_d \left(\mathrm{e}^{\frac{E_d - V}{\theta_d}} + 1 \right)^2}, \qquad \frac{\mathrm{d}^2 d_\infty}{\mathrm{d}V^2} = \frac{\mathrm{e}^{\frac{E_d - V}{\theta_d}} \left(\mathrm{e}^{\frac{E_d - V}{\theta_d}} - 1 \right)}{\theta_d^2 \left(\mathrm{e}^{\frac{E_d - V}{\theta_d}} + 1 \right)^3}. \tag{3.37}$$

The fact that the points must be in the critical manifold and that $\frac{\partial h}{\partial V} = 0$ gives the equalities and the fact that $\frac{\partial^2 h}{\partial V^2} \neq 0$ gives the non-equalities.

□

Since the critical manifold is easy to compute, the following result is fundamental to investigate the properties and location of the actual manifolds of interest in the actual system.

Proposition 3.11. *For $0 < \varepsilon \ll 1$, the critical manifold C_0 perturbs to invariant manifolds that exist in the full system and have the same topological structure and stability nature as C_0.*

Proof. These are consequences of Fenichel Theory. The limit $\varepsilon \to 0$ of the system with fast time derivatives in (3.29) gives orbits that converge during fast dynamics to solutions of the layer problem, whereas the limit $\varepsilon \to 0$ of the system with slow time derivatives in (3.29) gives orbits that converge during slow dynamics to solutions of the reduced problem. However, when we consider the full system with $\varepsilon \neq 0$, First Fenichel Theorem 2.3 guarantees the existence of locally invariant smooth manifolds diffeomorphic to a subset of the critical manifold, staying $O(\varepsilon)$ close and preserving stability properties.

□

3.2.3 Numerical Simulations

We will solve the system of differential equations numerically to illustrate the behavior of the variables, the nature of the solutions and the relevant objects in the dynamical system. The simulations will be carried out fixing $G_{Ca} = 0.025$ mS/cm^2 and varying G_K taking the values 0.05, 0.045 and 0.04 mS/cm^2. We will simulate 8000 milliseconds and we will choose the initial condition $(V(0), x(0), f(0)) = (0, 0.5, 0.5)$. For each case, we show trajectories in the state space and time series for each variable.

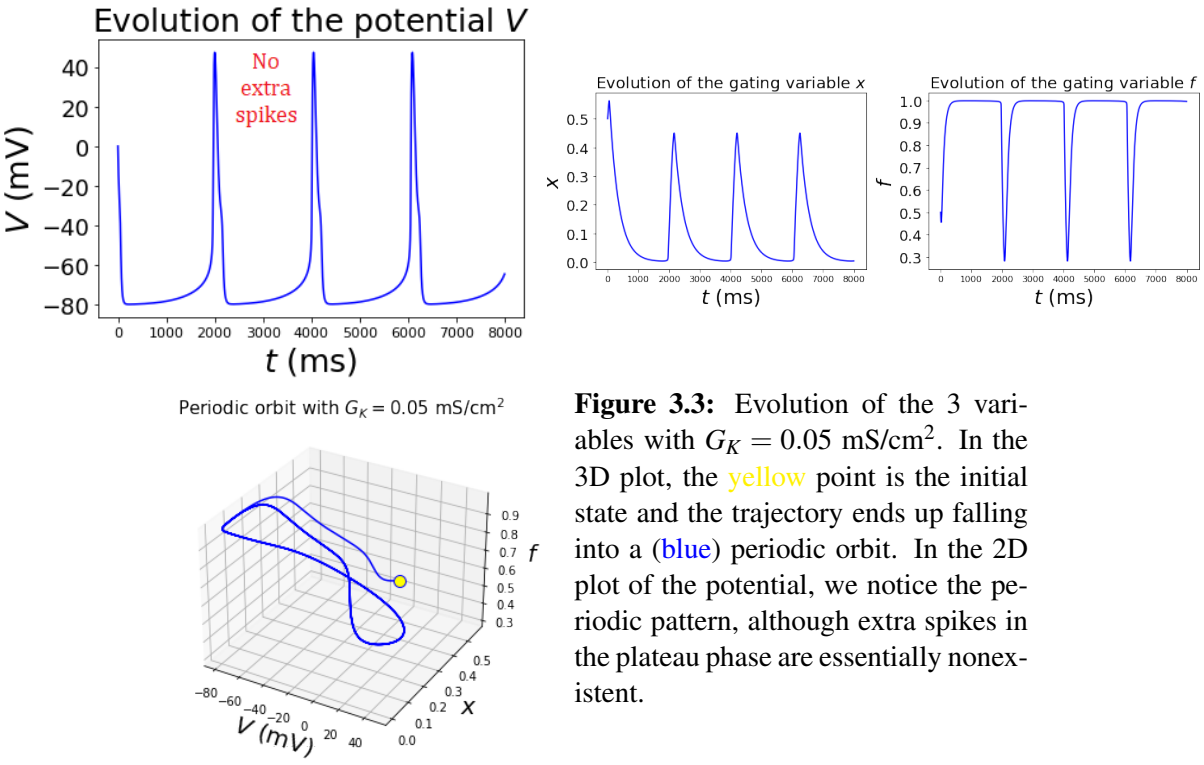


Figure 3.3: Evolution of the 3 variables with $G_K = 0.05$ mS/cm^2. In the 3D plot, the yellow point is the initial state and the trajectory ends up falling into a (blue) periodic orbit. In the 2D plot of the potential, we notice the periodic pattern, although extra spikes in the plateau phase are essentially nonexistent.

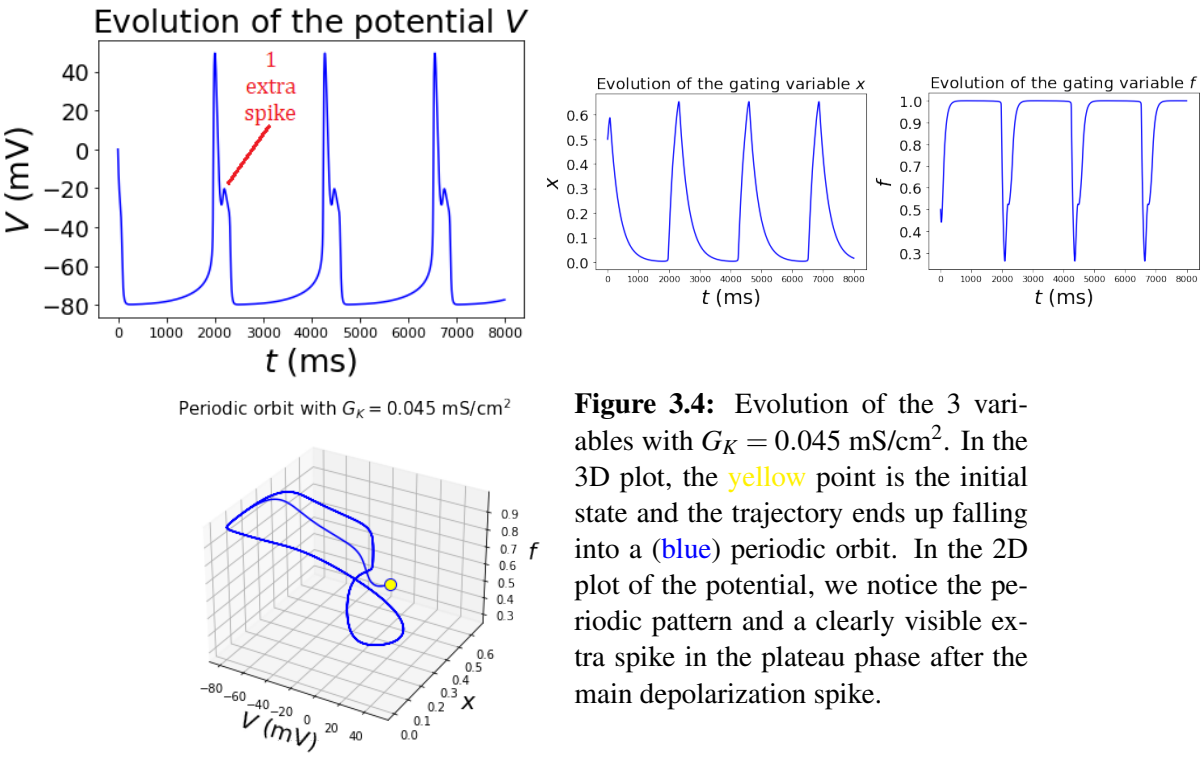


Figure 3.4: Evolution of the 3 variables with $G_K = 0.045$ mS/cm^2. In the 3D plot, the yellow point is the initial state and the trajectory ends up falling into a (blue) periodic orbit. In the 2D plot of the potential, we notice the periodic pattern and a clearly visible extra spike in the plateau phase after the main depolarization spike.

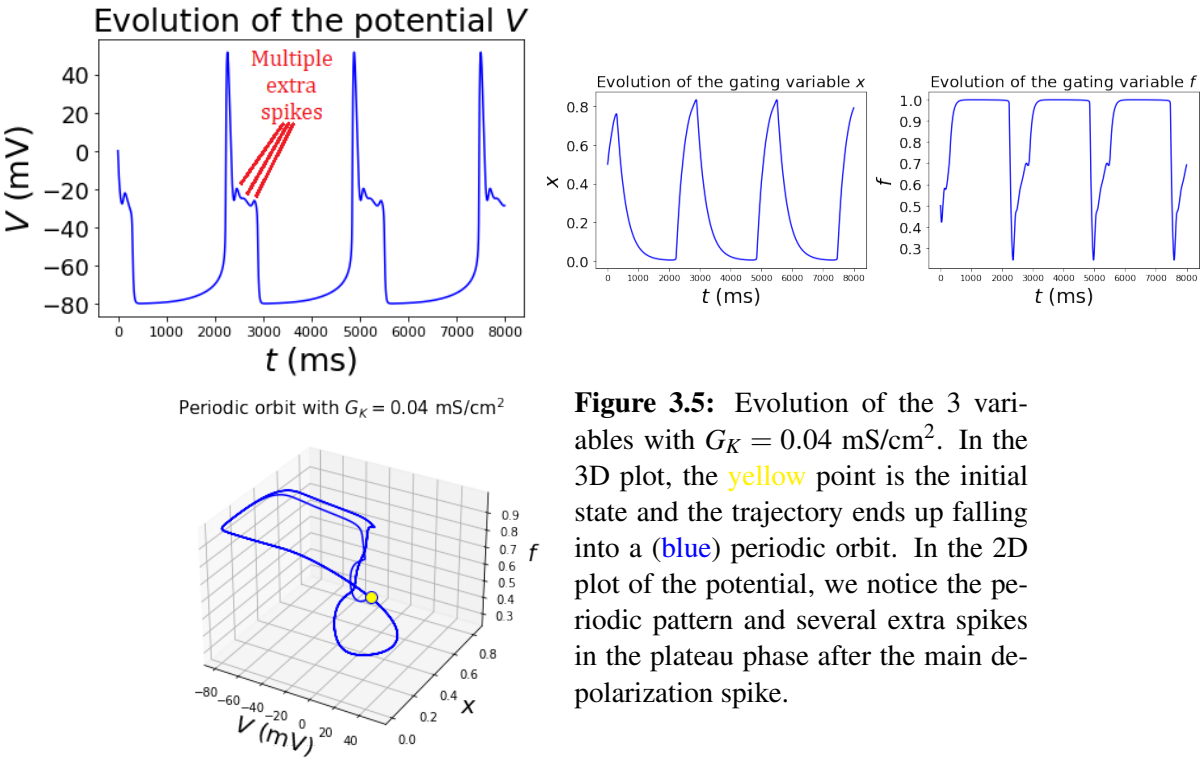


Figure 3.5: Evolution of the 3 variables with $G_K = 0.04$ mS/cm^2. In the 3D plot, the yellow point is the initial state and the trajectory ends up falling into a (blue) periodic orbit. In the 2D plot of the potential, we notice the periodic pattern and several extra spikes in the plateau phase after the main depolarization spike.

In any case, we see that the trajectories converge towards periodic orbits and the characteristic pulses of the action potential repeat periodically. Indeed, the time evolution of the transmembrane potential in this model reproduces correctly (at least, qualitatively) the expected behavior of voltage inside the cardiomyocytes with respect to the voltage outside the cardiomyocytes. That is, before a pulse, the cells initially present a clearly negative resting potential (in this case, we observe that it is around -80 mV) and they are polarized such that there is an excess of positive ions outside the cell membrane. Then, the stimulation of the cells makes them more permeable to positively charged ions that cross the membranes and the potential quickly turns positive. This rapid transition from highly negative to highly positive voltage is the depolarization process which leaves a narrow main spike in the potential graph after entering the plateau phase through the transient repolarization. In the plateau phase, the voltage will stay approximately constant. Here, we see a crucial difference depending on the conductance G_K chosen. For the highest conductance we have worked with ($G_K = 0.05$ mS/cm^2), the plateau phase is extremely short and there is practically no sufficiently long time interval where the voltage is maintained in a small range. As a result, there are no noticeable extra spikes in the plateau phase and this single spike indicates that this is the case of a normal heartbeat. However, decreasing G_K to 0.045 mS/cm^2 creates an additional bump in the plateau potential which is associated to an EAD. Setting G_K to 0.04 mS/cm^2 creates even more extra oscillations in the plateau potential and more extra spikes are seen, which leads to more EADs. In any case, as expected after the plateau, the repolarization occurs returning the potential to the negative resting value. Afterwards, the orbit will be traversed again and a similar action potential process will take place anew.

3.2.4 Relevant Points

Proposition 3.12. *The system (3.25) has a saddle equilibrium point for the parameter values* $G_K = 0.05,\ 0.045,\ 0.04$ *mS/cm*^{2}. *Additionally, there exists a folded-node point in one of the fold lines.*

Proof. We set the three equations equal to zero. From the second equation, it is possible to solve for $f = f_\infty(V)$ and from the third equation, it is possible to solve for $x = x_\infty(V)$. We substitute in the first equation to solve for V.

$$-(I_{Ca}+I_K)=0 \implies G_{Ca}\cdot\frac{1}{\mathrm{e}^{\frac{E_d-V}{\theta_d}}+1}\cdot\frac{1}{\mathrm{e}^{\frac{V-E_f}{\theta_f}}+1}\cdot(V-E_{Ca})+G_K\cdot\frac{1}{\mathrm{e}^{\frac{E_x-V}{\theta_x}}+1}\cdot(V-E_K)=0. \quad (3.38)$$

However, this time, the relevant equilibrium point of the system has to be computed numerically (with the numerical values presented in Table 3.1) because of the complexity of this first equation. For each value of G_K, we obtain an equilibrium point:

G_K (mS/cm^2)	V_{EQ}	f_{EQ}	x_{EQ}
0.05	-40.3817	0.9145	0.4809
0.045	-36.4343	0.8711	0.6711
0.04	-31.0882	0.7840	0.8560

Table 3.2: Equilibrium point (horizontal row) for each value of G_K.

The jacobian is

$$J(V,f,x)=\begin{pmatrix} -\frac{1}{C_m}\left[\frac{\partial I_{Ca}}{\partial V}+\frac{\partial I_K}{\partial V}\right] & -\frac{1}{C_m}\left[\frac{\partial I_{Ca}}{\partial f}+\frac{\partial I_K}{\partial f}\right] & -\frac{1}{C_m}\left[\frac{\partial I_{Ca}}{\partial x}+\frac{\partial I_K}{\partial x}\right] \\ \frac{f'_\infty}{\tau_f} & -\frac{1}{\tau_f} & 0 \\ \frac{x_\infty}{\tau_x} & 0 & -\frac{1}{\tau_x} \end{pmatrix}, \quad (3.39)$$

where

$$f'_\infty=-\frac{\mathrm{e}^{\frac{V-E_f}{\theta_f}}}{\theta_f\left(\mathrm{e}^{\frac{V-E_f}{\theta_f}}+1\right)^2}, \quad x'_\infty=\frac{\mathrm{e}^{\frac{E_x-V}{\theta_x}}}{\theta_x\left(\mathrm{e}^{\frac{E_x-V}{\theta_x}}+1\right)^2}, \quad d'_\infty=\frac{\mathrm{e}^{\frac{E_d-V}{\theta_d}}}{\theta_d\left(\mathrm{e}^{\frac{E_d-V}{\theta_d}}+1\right)^2}, \quad (3.40)$$

$$\begin{cases} \frac{\partial I_{Ca}}{\partial V}=G_{Ca}f\,[d'_\infty(V-E_{Ca})+d_\infty], \\ \frac{\partial I_K}{\partial V}=G_K x. \end{cases} \quad \begin{cases} \frac{\partial I_{Ca}}{\partial f}=G_{Ca}d_\infty(V-E_{Ca}), \\ \frac{\partial I_K}{\partial f}=0. \end{cases} \quad \begin{cases} \frac{\partial I_{Ca}}{\partial x}=0, \\ \frac{\partial I_K}{\partial x}=G_K(V-E_K). \end{cases} \quad (3.41)$$

Substituting each equilibrium point with each G_K, the corresponding jacobian has one negative eigenvalue and two positive eigenvalues. Thus, in any of the three cases, the equilibrium is a saddle point.
A folded-node point (folded singularity) is an equilibrium point of the so-called *desingularized system*, but not of the slow subsystem. The desingularized system is built by expressing the slow flow in terms of g_1, g_2 and partial derivatives of h and using a rescaled time t_d with a factor $-\left(\frac{\partial h}{\partial V}\right)^{-1}$:

$$\begin{cases} \frac{\mathrm{d}V}{\mathrm{d}t_d}=\frac{\partial h}{\partial f}g_1+\frac{\partial h}{\partial x}g_2, \\ \frac{\mathrm{d}x}{\mathrm{d}t_d}=-\frac{\partial h}{\partial V}g_2. \end{cases} \quad (3.42)$$

If the second equation vanishes, the point must be in the fold lines obtained in Proposition 3.10. Joining the fact that the first equation also vanishes and substituting the parameter values of Table 3.1, we solve numerically the following system for each G_K:

$$\begin{cases} G_{Ca}\frac{1}{\mathrm{e}^{\frac{E_d-V}{\theta_d}}+1}(V-E_{Ca})\frac{\frac{1}{\mathrm{e}^{\frac{V-E_f}{\theta_f}}+1}-f}{\tau_f}+G_K(V-E_K)\frac{\frac{1}{\mathrm{e}^{\frac{E_x-V}{\theta_x}}+1}-x}{\tau_x}=0, \quad \text{(desingularization)} \\ f=-\frac{G_K x(V-E_K)}{G_{Ca}(V-E_{Ca})}\left(\mathrm{e}^{\frac{E_d-V}{\theta_d}}+1\right)=-\frac{G_K x}{G_{Ca}\left(\frac{\mathrm{e}^{\frac{E_d-V}{\theta_d}}}{\theta_d\left(\mathrm{e}^{\frac{E_d-V}{\theta_d}}+1\right)^2}(V-E_{Ca})+\frac{1}{\mathrm{e}^{\frac{E_d-V}{\theta_d}}+1}\right)}. \quad \text{(fold line)} \end{cases} \quad (3.43)$$

G_K (mS/cm^2)	V_{FN}	f_{FN}	x_{FN}
0.05	−24.7923	0.5001	0.4731
0.045	−24.7923	0.5368	0.5642
0.04	−24.7923	0.5735	0.6781

Table 3.3: Folded-node point (horizontal row) for each value of G_K.

□

3.2.5 Dynamical Interpretation

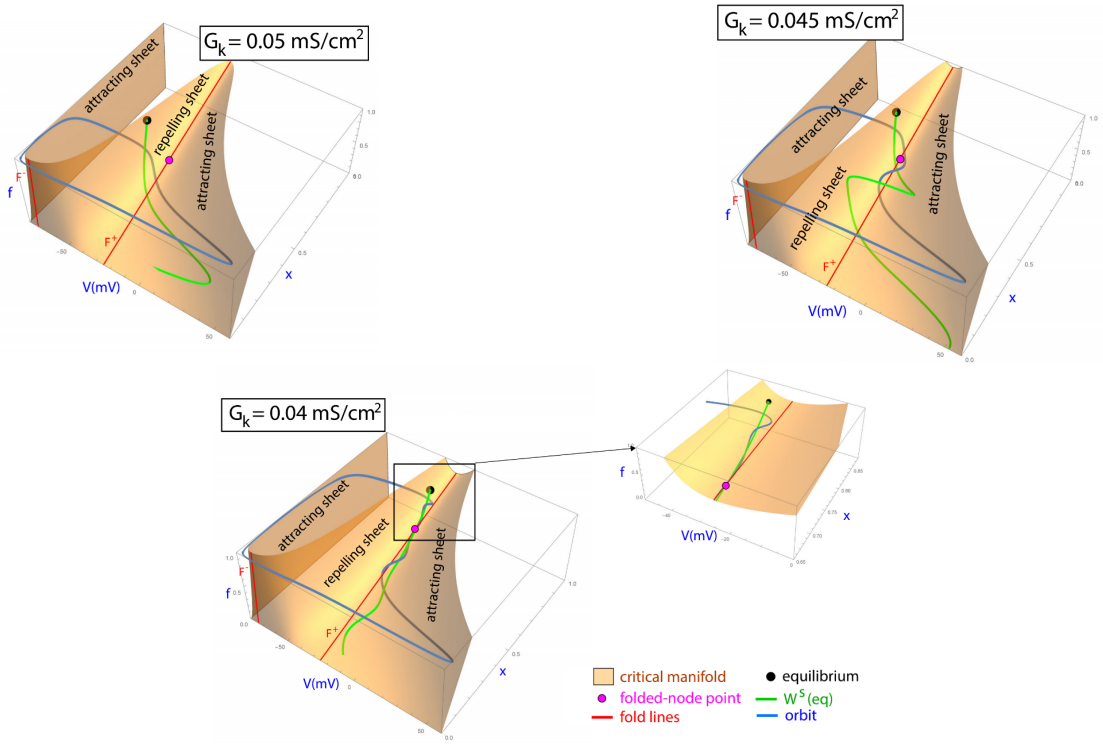


Figure 3.6: Relevant dynamical objects for $G_K = 0.05,\ 0.045,\ 0.04$ mS/cm^2 (extracted from [2]).

We see how the variation of G_K shifts the equilibrium and folded-node points and changes the shape of the periodic orbit. In the case with $G_K = 0.05$ mS/cm^2, the orbit closely traverses the critical manifold evolving between the two attracting sheets. When the conductance is lowered to $G_K = 0.045$ mS/cm^2, one EAD appears in the time series. The orbit is also slightly extended in the direction of the axis associated to the variable x. Now, the orbit passes near the folded-node point and the orbit makes an extra loop on the repelling sheet before evolving towards the attracting sheet on the other side. For the even lower value of $G_K = 0.04$ mS/cm^2, more loops are observed with quick oscillations near the folded-node point. The orbit evolves around the stable manifold associated to the equilibrium and transitions between the repelling and an attracting sheet before getting near the equilibrium. Finally, after this short phase, the orbit is repelled from the repelling sheet and attracted to the other attracting sheet.
This behavior with small oscillations in the potential is connected to the EADs observed in the plateau phase. Furthermore, the different number of loops is also connected to the specific number of EADs. In fact, we have just observed the canard phenomenon. No maximal canards are shown for the first value of G_K. For the next lower value, the dynamics fits the definition of a maximal canard in Definition 2.14, with the orbit staying close to an attracting sheet of the critical manifold (lying in a slow attracting manifold), passing near a point where normal hyperbolicity is lost and staying close to a repelling sheet (lying in a slow repelling manifold) for some time. For even lower values, more EADs are generated by more canards.

Chapter 4
Conclusion and further work

The development and analysis of models of differential equations allows us to extract information about the evolution of a set of mutually dependent variables. In this thesis, a concise summary of dynamical systems theory was presented with a focus on the kind of objects which are usually studied in fast-slow systems. Then, we discussed some classical models, the physical motivation and consequences of their particular equations and their dynamical and numerical properties.
In the case of the Karma Model, we studied the dynamics of the pure ODE version. Three types of equilibria are present and we observed the critical manifold, the shape of the orbits and a heteroclinic connection. Out of the singular limit, the former are perturbed to nearby manifolds with Fenichel Theorems and the latter exists in the actual system as seen with the Poincaré-Bendixson Theorem.
In the case of the Reduced Luo-Rudy Model, we successfully observed how the full action potential process takes place in its depolarization, plateau and repolarization phases. We also observed the mechanism of generation of EADs leading to arrhythmias depending on the parameters and how the physical interpretation is related to the observation of the critical manifold, canards, etc.
For further research, the effect of different combinations of parameters and a deep analysis of the dynamical effect of the addition of non-negligible external currents could be studied.
Since the last century, great efforts have been made to encapsulate the most important features of cardiomyocyte excitability through numerous models. An extensive list of up to 47 models of cardiac cells with their distinctive features can be found in [8]. A model can be as complex as its designer desires depending on the number of differential equations and parameters. Some simple models with a reduced number of equations and parameters may be a reasonable approximation of the qualitative properties that show up in reality. This does not mean that they are less relevant since understanding their features is fundamental to comprehend the characteristics of more complex systems and to know how to build them correctly. In recent years, new articles and results on some of these classical simple models are still being published (see [2] and [9] in relation to the models discussed in this thesis).
However, recent technological advances allow the treatment of increasingly complex systems to explain very accurately some specific phenomena. To cite an example: let us have a look at the highly complex *Sato Model* with recent research on it (see [3]). It is defined by 27 variables and 177 parameters. The analysis of the model requires sophisticated computational techniques to explore the existence of bifurcations, chaotic attractors or canards. For instance, in order to detect accurately the presence of EADs leading to arrhythmias, we need biophysically detailed models. The high number of parameters allows to adjust the evolution of the variables according to the experimental data. The high number of variables enables the precise modeling of the evolution of biophysical magnitudes. In the aforementioned model, we consider ionic currents for Ca^{2+}, Na^{+} and K^{+} (up to 9 different currents including separated fast and slow components, inward and outward currents and exchange currents between different ions). Among the 27 variables, we have 10 gating variables and separate variables for the transmembrane potential, intracellular sodium concentration and several types of calcium concentrations and related currents.
This clearly illustrates the thoroughness with which the most accurate models are built. Further research in this area should be directed to the study of this kind of complex models. They are not always easily tractable, but they are the key to understand the actual behavior of cardiac cells, which are, in nature, a really complex biological system. There may be interactions between cells or more advanced physical effects that realistic models should take into account. The analysis of the properties of highly realistic dynamical systems can lead to the development of meaningful interpretations of the physical phenomena in a mammal's heart. In the case of the presence of problems like arrhythmias, an accurate mathematical study allows to understand the precise chemical treatments that should be needed and the specific regions of the parameter and state spaces where special phenomena occur. This creates a link between mathematics and other sciences, demonstrating the potential of mathematics as an applied discipline.

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